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Homo sapiens Is as *Homo sapiens* Was

Behavioral Variability versus "Behavioral Modernity" in Paleolithic Archaeology

by John J. Shea

Paleolithic archaeologists conceptualize the uniqueness of *Homo sapiens* in terms of "behavioral modernity," a quality often conflated with behavioral variability. The former is qualitative, essentialist, and a historical artifact of the European origins of Paleolithic research. The latter is a quantitative, statistically variable property of all human behavior, not just that of Ice Age Europeans. As an analytical construct, behavioral modernity is deeply flawed at all epistemological levels. This paper outlines the shortcomings of behavioral modernity and instead proposes a research agenda focused on the strategic sources of human behavioral variability. Using data from later Middle Pleistocene archaeological sites in East Africa, this paper tests and falsifies the core assumption of the behavioral-modernity concept—the belief that there were significant differences in behavioral variability between the oldest *H. sapiens* and populations younger than 50 kya. It concludes that behavioral modernity and allied concepts have no further value to human origins research. Research focused on the strategic underpinnings of human behavioral variability will move Paleolithic archaeology closer to a more productive integration with other behavioral sciences.

How we humans differ from other living things is the central question of anthropology. All of the research questions with which anthropologists concern themselves ultimately lead back to this one. It is the reason why anthropology and allied fields exist separately from other fields of animal behavior and vertebrate paleontology. In paleoanthropology, investigations of human uniqueness today are part of a larger inquiry into the origins of modern humans. This research unites two somewhat separate groups of questions, those about the physical evolution of *Homo sapiens* and those concerning the evolution of human behavior.

Fossil and molecular evidence point unambiguously to our species' first appearance in Africa by at least 200 kya and broad distribution within that continent by 100 kya (Bräuer 2008; Grine et al. 2007; McDougall, Brown, and Fleagle 2005; Pearson 2008; Rightmire 2008; Trinkaus 2005; Weaver and Roseman 2008). Some human populations began dispersing from Africa by at least 100–40 kya, reaching southeast Australia by at least 40 kya (O'Connell and Allen 2004). By 12 kya *H. sapiens* was in the Americas (Waguespack 2007) and was present on all major continental land masses except Antarctica. This dispersal appears to have been facilitated by a suite of unique or "modern" behavioral strategies that enabled our species to construct uniquely broad, flexible, and stable ecological niches (Marean 2007; O'Connell 2006). Evidence for this unique human niche appears episodically in African contexts before 50 kya, then consistently both within and outside of Africa after 40–50 kya (Klein 2008; McBrearty 2007; Stiner and Kuhn 2006). Though some researchers see the achievement of human "behavioral modernity" as a saltational event, a prehistoric "revolution" (Bar-Yosef 2002; Klein 2008; Mellars 2007), others envision it as a more gradual, sometimes even recursive process of cumulative behavior change (Gamble 2007; McBrearty and Brooks 2000; references in Hovers and Kuhn 2006 and Mellars et al. 2007).

Questions about fossils and behavior are both problematic, but in different ways. Life history and taphonomy influence fossil morphology, but their ultimate sources of variation are Darwinian evolutionary processes. Individual researchers may disagree whether a particular morphology is present on a fossil or about the "polarity" of that morphology (whether it is primitive or derived relative to other morphologies). Nevertheless, fossils with similar derived morphologies are inferred to be more closely related evolutionarily than fossils between which there are contrasting derived morphologies (Lieberman 1995). A hominin fossil with a mental eminence, divided superciliary arch, and canine fossa is more closely related to living *H. sapiens* than is one that lacks these derived features (Tattersall and Schwartz 2008).

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In researching the evolution of human behavior, Paleolithic archaeologists face a very different set of problems. Variation among artifacts and ecofacts results from variability in behavior. Behavior and its tangible by-products "evolve" only in a metaphorical sense. The actions involved in making a ceramic pot are neither more nor less "derived" than those involved in crafting a reed basket. Both activities and their by-products reflect correspondingly different strategic costs and benefits in container design. Their occurrence in the ethnographic record reflects strategic factors (e.g., time budgeting, transportability, durability, resistance to thermal damage), and uniformitarian theory suggests strategic factors influenced their distribution in the past as well. Unlike the shared-derived features of fossils, similar archaeological associations do not necessarily indicate a close evolutionary relationship. Living humans who create replicas of Lower Paleolithic tools, either deliberately (in the case of flint knappers) or de facto (in countless ethnographic contexts), are not evolutionarily "closer" to *Homo erectus* than the rest of us are. Similarities between the archaeological traces of human behavior separated by tens or even hundreds of thousands of years can arise from convergence in ways that are vastly more difficult to detect than morphological convergence because of homoplasy (Clark and Riel-Salvatore 2006; Shea 2006*a*). This basic difference in the nature of physical anthropological and archaeological evidence is a source of many problems in current research on the origins and evolution of *H. sapiens*.

Our species differs from our nearest primate relatives in our capacity for behavioral variability (Potts 1998). Except where constrained by physics, chemistry, or biology, everything humans do we do in more than one recognizably different way. Paleolithic archaeologists have developed a habit of conflating "behavioral variability" with "behavioral modernity" and "behavioral complexity." Behavioral variability and behavioral modernity are very different concepts (table 1). Variability is a measurable quality of all human behavior expressed in term of modality, variance, skew, and other quantitative/statistical properties. These qualities change through time and space, and they do not necessarily follow a preferred direction. Trends are recognizable only in hindsight, ex post facto. A more versatile individual (or species) can become more specialized, or vice versa, in response to variation in selective pressures. All living humans, extinct humans, and extinct hominins are assumed to have possessed a capacity for behavioral variability at least comparable in scale to living African apes. We do not know the antiquity of our species' current degree of behavioral variability, but there are only three possibilities: (1) it evolved after our species' origin and is a characteristic of only some *H. sapiens*, (2) it evolved at the same time our species split from ancestral hominins and is a species-specific characteristic, and (3) it evolved before *H. sapiens*' origin and is a characteristic shared by more than one hominin species. (In theory, "behavioral complexity" is a particular dimension of behavioral variability, but in practice, archaeologists use this term synonymously with behavioral modernity [e.g., Mellars 2007].)

Behavioral modernity is a condition characterized qualitatively in terms of the inferred presence or absence of particular "modern human behaviors." For historical reasons (discussed below) these behaviors are usually those associated with the European Upper Paleolithic. All living humans are assumed to be behaviorally modern, but archaeologists disagree over the extent to which behavioral modernity can be extrapolated to extinct human and hominin populations. Living panins (chimpanzees and bonobos) are not behaviorally modern. Behavioral modernity implicitly assumes there is a trend to behavioral change in *H. sapiens* evolution, one in which earlier "behaviorally archaic" humans were transformed into more versatile "behaviorally modern" ones. This trend is nonreversible. Neither individuals nor populations can become "more archaic"/"less modern." Trend models of evolution have a long history of being wrong (Gould 1987). In this paper, I argue that the hypothetical evolutionary trend that is the core assumption of the behavioral modernity model in Paleolithic archaeology is wrong, too.

The assumptions we make about the behavior of the earlier *H. sapiens* populations have to be grounded in uniformitarian theoretical principles. All living humans are capable of wide behavioral variability. There is no evidence that such skeletal contrasts as exist between the earliest *H. sapiens* and those reading this paper indicate significant differences in the capacity for behavioral variability. Yet current models for the

Table 1. Differences between behavioral variability and behavioral modernity

Variable	Behavioral variability	Behavioral modernity
Defining characteristics	All human behavior	Derived features of the European Upper Paleolithic
Scale of variation	Quantitative (statistical)	Qualitative (presence/absence)
Diachronic trend	Possible, but not necessary (recognizable in hindsight only)	Cumulative and linear
Trends reversible	Yes	No
Shared by all living humans	Yes	Yes
Shared by all extinct *Homo sapiens*	Yes	No
Shared by hominins other than *H. sapiens*?	Yes, though differences likely	No
Shared by hominins and panins (chimpanzees, bonobos)	Yes, though differences demonstrable	No

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evolution of modern human behavior assume that the earliest *H. sapiens* were less capable of behavioral variability than we are today. No serious scientific investigation can begin by assuming that the null hypothesis is wrong. If the null hypothesis is correct (or if we cannot falsify it), then searching for the origins of modern human behavior in the archaeological record of *H. sapiens* diverts scarce time, energy, and research funds away from more productive lines of inquiry into the evolution of human behavioral variability.

In this paper, I make three arguments. The first is that both behavioral modernity and modern human behavior are archaeological constructs originally developed in European Paleolithic research that have been projected to a global scale without sufficient theoretical justification. The belief that the achievement of behavioral modernity marks a milestone in human evolution also derives from a long-standing tradition of using narrative explanatory models in paleoanthropology.

The second argument is that archaeological evidence associated with later Middle Pleistocene *H. sapiens* in East Africa indicates a capacity for behavioral variability equivalent to that associated with Late Pleistocene and Early Holocene humans.

The third argument is that Paleolithic archaeologists' search for a derived and unique pattern of *H. sapiens* behavior will be far more successful if we think about behavior strategically, in terms of costs and benefits imposed under situationally variable circumstances. This kind of rethinking will lead us in a more productive direction, to testable propositions about the sources of our species' capacity for behavioral variability.

What Is Behavioral Modernity?

Archaeological Definitions of Behavioral Modernity

Strictly speaking, behavioral modernity and modern human behavior refer to two different things—the former a quality of behavior and the latter the distinctive by-products of that behavior. In practice, Paleolithic archaeologists use the terms as synonyms. Because Paleolithic archaeology has deeper historical roots in Europe than in other regions, current definitions of modern human behavior/behavioral modernity hew closely to the ways in which the European Upper Paleolithic archaeological record differs from the European Middle Paleolithic archaeological record (table 2; for a detailed review and references, see Henshilwood and Marean 2003). Most of these modern human behaviors share the following characteristics: (1) there is durable archaeological evidence for them, (2) they occur among ethnographic hunter-gatherers, and (3) they are not consistently associated with other hominin species. Much current debate in Paleolithic archaeology involves efforts to reconcile this definition of behavioral modernity based on *Homo sapiens*' European Upper Paleolithic archaeological record with evidence from other regions. The archaeological records associated with *H. sapiens* in Southwest Asia, Africa, and elsewhere do not feature identical sets of contrasts between prehistoric periods analogous to the European Middle and Upper Paleolithic. In Africa and Southwest Asia, *H. sapiens* fossils and those immediate precursor taxa are associated with Lower and Middle Paleolithic archaeological assemblages (McBrearty and Brooks 2000; Shea 2003*a*). A sparse fossil record for mainland South Asia makes the antiquity and archaeological associations of early *H. sapiens* there largely matters for conjecture (Petraglia and Allchin 2007). The archaeological associations of the Pleistocene *H. sapiens* population of Sahul (New Guinea and Australia) defy simple classification as Lower, Middle, or Upper Paleolithic (Holdaway and Stern 2004; Mulvaney and Kamminga 1999).

Some modern behaviors previously thought to be associated solely with *H. sapiens* have been detected in contexts referable to Neanderthals (d'Errico 2003; Zilhão 2007). Neanderthals and their associated archaeological record are interesting in their own right, of course, and central to European prehistory (Mellars 1996). Yet fossil morphology combined with studies of DNA both from fossils and from living humans now shows that Neanderthals and their associated archaeological record are largely irrelevant for models concerning early phases of *H. sapiens* evolution (Harvati, Frost, and McNulty 2004; Serre and Pääbo 2006).

Early African Homo sapiens *and Behavioral Modernity*

While it is possible that behavioral modernity first emerged on the expanding frontier of our species' geographic range in Eurasia (Lewis-Williams 2002; Shea 2003*b*), most recent research on this subject focuses on Africa. Among the most commonly cited theories about the evolution of modern human behavior in Africa, there are considerable differences about the inferred mode and tempo of this evolutionary process.

Klein (1992, 2008) argues that differences between the behavior of African Middle Stone Age humans on the one hand and both African Later Stone Age and Eurasian Upper Paleolithic humans on the other resulted from the spread of a neural mutation linked to fully facultative language and cultural behavior around 50 kya. The main evidence for this change is improvements in foraging efficiency and technological complexity inferred from the zooarchaeological and lithic records. The main weakness of the "neural mutation" hypothesis is that the hypothetical neural mutation does not leave a clearly detectable signal in the fossil record. Archaeological tests of this hypothesis require one to accept that abilities conferred by this mutation are responsible for particular patterns of *H. sapiens* behavior. This does not make the neural mutation hypothesis untestable, merely difficult to test with archaeological or fossil evidence.

For a time, a gene polymorphism linked to speech, FOXP2,

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Table 2. Presence/absence contrasts for behaviors indicated for Middle Paleolithic (MP) and Upper Paleolithic (UP) periods in Europe and Southwest Asia and the African Middle and Later Stone Ages (MSA, LSA)

Behavior	Europe MP	Europe UP	SW Asia MP	SW Asia UP	Africa MSA	Africa LSA
Persistent occupation of harsh environments	. . .	+	+?	+	+	+
Seasonal mobility strategies	+?	+	+	+	+	+
Effective hunting of large mammals	+	+	+	+	+	+
Systematic fishing and exploitation of shellfish and aquatic mammals and birds	+?	+	. . .	+?	+	+
Organized use of domestic space	. . .	+	+	+	?	+
Patterned artifact diversity/standardization of artifact types	. . .	+	. . .	+	+	+
Systematic prismatic blade production	. . .	+	+	+	+	+
Carved bone, antler, ivory, and stone artifacts	. . .	+	. . .	+	+	+
Symbolic use of mineral pigments	+?	+	+	+	+	+
Personal adornments	. . .	+	+	+	+	+
Representative and abstract art	. . .	+	. . .	+	. . .	+
Ritualized symbolic behaviors (mortuary practices)	+?	+	+	+	+?	+
Musical instruments	. . .	+	. . .	. . .	. . .	+
Long-distance exchange networks	. . .	+	. . .	+?	+	+

Sources. For detailed recent reviews of this evidence and sources, see for Europe, Conard (2007), d'Errico (2003), d'Errico et al. (2003), Gamble (1999), and Mellars (1994). For Southwest Asia, see Bar-Yosef (2002) and Shea (2007*a*). For Africa, see Barham and Mitchell (2008), McBrearty (2007), McBrearty and Brooks (2000), and Willoughby (2007).
Note. Question marks (?) indicate that evidence is inconclusive.

was thought to be a likely candidate for this mutation (Enard et al. 2002; Klein and Edgar 2002), but this polymorphism has since been detected in Neanderthal DNA (Krause et al. 2007). This suggests FOXP2 was present in Neanderthals' and *H. sapiens*' Middle Pleistocene common ancestor. Though some have accepted this finding as a falsification of the neural mutation hypothesis, it must be kept in mind that neurogenetics is a young science, and paleogenetics is younger still. Countless discoveries about the genetic basis of human and hominin behavioral variability will be made in the very near future. One cannot equate prospect with proof, but the fate of the neural mutation hypothesis ought not to be conflated with the fate of the FOXP2 gene as the arbiter of human behavioral modernity. As matters stand today, the presence of FOXP2 in Neanderthal DNA suggests there was strong selective pressure for speech and, by implication, for language too among the Middle Pleistocene hominin populations ancestral to both Neanderthals and *H. sapiens*.

Ambrose (1998) has argued that a sharp drop in global temperatures caused by the Mount Toba supereruption ca. 70 kya drastically reduced African human populations and concentrated the survivors into refugia where modern human behavior evolved through a series of correlated socioecological changes necessary for survival. These changes included technological improvements (Later Stone Age lithic technology) and the formation of extensive social networks reinforced by patterns of long-distance raw material exchange (Ambrose's "troop-to-tribe transition"). This hypothesis has gained support from recent evidence for recurrent megadroughts in sub-Saharan Africa during late Middle and Later Pleistocene times (Scholz et al. 2007).

Criticisms of this bottleneck/refugium hypothesis have challenged the suggested severity of the Mount Toba supereruption for global climate (Gathorne-Hardy and Harcourt-Smith 2003) and the theoretical basis for inferring demographic bottlenecks (as opposed to persistently small population sizes; Premo and Hublin 2009; Weaver and Roseman 2008), as well as the historical reality of the proposed African climatic refugia (Cowling et al. 2008). South Africa and East Africa are currently the leading contenders for the specific location of these hypothetical refugia (Basell 2008; Brandt 2006; Deacon and Deacon 1999). The woodlands of West Africa, the Atlantic watershed of the Atlas Mountains, and the coastlines of the Mediterranean Sea and the Red Sea also have their exponents among prehistorians. Migration is a difficult thing to prove in prehistory (Clark 1994). Thus far, most archaeological "proofs" that one or another of these regions was the *Urheimat* (original homeland) of our species consist of little more than maps showing arrows to indicate conjectural human migrations, much as they did among earlier migration-driven models of human prehistory (Kossina 1911).

McBrearty and Brooks (2000; McBrearty 2007) have proposed a model in which modern behaviors developed cumulatively in Africa over the course of the last 280 kya. (Sometimes mischaracterized as a "gradualistic" model, McBrearty and Brooks actually left the question of rates of change an open one.) They explain this pattern largely in terms of pop-

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ulation-density-dependent effects, such as subsistence intensification and the extension of social networks in response to environmental deterioration (McBrearty and Brooks 2000: 532). "Population packing" is demonstrably a factor in the technological complexity and subsistence intensification of recent human hunter-gatherers (Binford 2001); thus, the middle-range theoretical basis for their hypothesis is credible.

One argument that can be made against this model is that the appearance of a cumulative process could be an artifact of methodology. McBrearty and Brooks's model treats the oldest occurrence of evidence for any particular modern human behavior anywhere in Africa as evidence for the capacity for that behavior everywhere in Africa from that point onward. Recent contexts are always more abundant than ancient ones simply as a result of geological attrition, and larger samples of any heterogeneous data set always show greater variability. The apparent accumulation of more "modern" behaviors in more recent phases of African Middle Stone Age prehistory may simply reflect the fact that there are more archaeological samples from recent time ranges, and thus a more accurate reflection of actual human behavioral variability during those periods than in earlier ones. If early (i.e., later Middle Pleistocene African) *H. sapiens* already possessed the full suite of modern human behavior, site taphonomy and the method McBrearty and Brooks deploy to record the presence of those behaviors would create the illusion of a cumulative trend.

In reviewing recent theories about the origin of modern human behavior, Henshilwood and Marean (2003) conclude that the only kind of behavior that meets the criteria of preservation, clarity, and theoretical justification is symbolic behavior. They cite durable evidence for symbolic behavior in the form of mechanically modified mineral pigments (red ochre) and perforated shells found in contexts dated to 74–84 kya at Blombos Cave in South Africa as early evidence for this symbolic capacity (Henshilwood et al. 2001). This evidence, recently augmented by finds of red ochre dating to 164 kya at Pinnacle Point Cave (Marean et al. 2007), seems to suggest a connection between durable evidence of symbolic behavior and early human adaptations to coastal environments in southern Africa.

Parsing behavioral modernity to symbolic behavior alone creates problems for archaeological hypothesis testing, specifically the problem of "false negative" evidence. Recent humans do make extensive use of durable media for the production of exosomatic symbolic artifacts. Thus, finding ochre-stained tools, perforated shells, or rock art is clear and convincing evidence for human symbolic behavior. But the absence of such evidence does not necessarily mean the originating symbolic capacity was absent. Recent humans make extensive use of perishable media as well. In those parts of East Africa where the earliest *H. sapiens* fossils have been recovered, ethnographic humans use many forms of exosomatic symbols made of durable materials (Hodder 1982), but they also use media that would be impossible to detect in an archaeological record a few centuries in age, much less one 200,000 years old. These practices include body painting (with red ochre, chalk, and various colored sediments), scarification, personal adornment with perishable objects (colored clothing, strings, wooden beads, labrets, and earplugs), and artifact stylistic variability (different kinds of baskets, wooden clubs/walking sticks). Can we be sure our ancestors were not doing broadly similar things? Absence of unambiguous exosomatic symbols in the archaeological record may not so much reflect the absence of symbolic behavior but rather the presence of weak incentives or disincentives for crafting exosomatic symbols out of durable media. Inasmuch as exosomatic symbol use is a universal among all known *H. sapiens* populations, preservation bias has to be considered a simpler explanation for the absence of symbolic artifacts than the actual absence of symbolic behavior. Regional differences in archaeological recovery techniques, particularly the use of wet sieving, may also account for differences in the recovery of small symbolic artifacts such as beads. Occurrences of mineral pigments and personal adornments in Neanderthal contexts in Europe suggest that whatever symbolic capacities underlie the production and use of these artifacts, they were shared by at least two Late Pleistocene hominin species.

This brief review does not exhaust the list of recent theories about the evolution of modern human behavior in Africa (Clark 1989; Deacon and Deacon 1999; Foley and Lahr 1997; Kingdon 1993; Kusimba 2003; Lahr and Foley 1994; Stringer and McKie 1996; Willoughby 2007). Adding more theories, however, does not change the overall pattern in the evidence. Current models for the origins of modern human behavior in Africa differ from one another over mode and tempo of this evolutionary change, but there is virtual unanimity over just which behaviors make up behavioral modernity and over the assumption that recent and ancient African *H. sapiens* differed in the performance of those behaviors.

That the behaviors in question are those that distinguish the European Upper Paleolithic from the European Middle Paleolithic is a strange set of affairs. Africanist prehistorians are acutely aware of the European origins of these working definitions, and most agree that they are inappropriate (Henshilwood and Marean 2003; McBrearty 2007; McBrearty and Brooks 2000; Wadley 2001), yet these definitions have neither been abandoned nor replaced by substantively different ones (Barham and Mitchell 2008; Willoughby 2007:324). The absence of modern behaviors unique to Pleistocene Africa is even more odd. For at least 195,000 years, *H. sapiens* was an African hominin, and for much of that period (>50–60 kya), it was exclusively African. There must have been behaviors derived uniquely among African *H. sapiens*. And yet, to read much of what has been written recently about *H. sapiens* evolution on that continent, the only such uniquely African pattern of human behavior appears to be the capacity to persist successfully for long periods of geological time without acting like Upper Paleolithic Europeans.

Nothing in recent paleoanthropological research challenges

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the belief all living humans are equally divergent from hominin ancestors and therefore equally behaviorally modern. There is also a consensus that all humans (i.e., *H. sapiens*) who have lived since Pleistocene times have been behaviorally modern. The archaeological consensus on human behavioral modernity deteriorates as one looks farther back into the Late Pleistocene record, beyond 40–50 kya. Some components of modern human behavior appear before the oldest-known dates for *H. sapiens* fossils, but there appears to be a consensus that behavioral modernity evolved during the course of *H. sapiens*' evolution. Evidence presented in this paper will show this consensus is wrong and that the capacities for behavioral variability underwriting what we call behavioral modernity or modern human behavior are at least as old as the oldest skeletally modern-looking *H. sapiens*. The problem is that we can neither detect such behavioral variability nor test hypotheses about it by continuing to look for behavioral modernity or modern human behavior in the same way as we have been doing for the last 30 years.

Why Is "Behavioral Modernity" Defined the Way It Is?

That there are such concepts as "behavioral modernity" and "modern human behavior" in human origins research reflects the historically contingent priority of Paleolithic archaeology in Europe and a tradition of using narrative explanatory models in paleoanthropology.

Extrapolation from European Evidence

"Anchoring" is a psychological phenomenon in which the first observations in a series disproportionately influence the perception of subsequent ones (Tversky and Kahneman 1974). A similar phenomenon lies at the heart of the behavioral modernity concept. Paleolithic archaeology began in Europe (Grayson 1983), a region with a uniquely rich Upper Paleolithic record (Gamble 1999). It was (and remains) a logistically convenient place to conduct fieldwork for scholars based in European and American institutions. Such researchers make up the overwhelming majority of professional Paleolithic archaeologists and hominin paleontologists. European chronological priority has meant that for much of the twentieth century, archaeologists framed general models of human behavioral evolution by extrapolating from the European evidence. *Homo sapiens*' quintessential derived behavioral characteristics were those seen in Upper Paleolithic archaeological assemblages associated with the oldest European *H. sapiens* fossils. These behaviors were usually presented in terms of contrasts with the Middle Paleolithic evidence associated with European Neanderthals (Dennell 1983; Mellars 1973).

Paleolithic archaeologists' persistent fascination with prismatic blades is a clear example of this extrapolation from European evidence (Bar-Yosef and Kuhn 1999). In Europe, evidence for the systematic production of prismatic blades coincides with the first appearance of *H. sapiens* in Upper Paleolithic contexts. For decades, Paleolithic archaeologists have routinely inferred a causal connection from this association. Discoveries of prismatic blade production in Middle or Lower Paleolithic contexts in Asia, Africa, or even Europe have been seen as evidence for "precocious" behavioral modernity (Garrod 1962; Wadley 2001). Yet it has never been shown that blade production calls for any greater measure of manual skill or intelligence than any of the other flint-knapping strategies practiced by Middle Pleistocene *Homo*. Indeed, as Eren, Greenspan, and Sampson (2008) have demonstrated, blades do not, as commonly supposed, necessarily yield more cutting edge per unit mass of stone than other core-reduction methods. To paraphrase Freud on cigars, sometimes a blade is just a blade.

Finds of *H. sapiens* fossils in Levantine Middle Paleolithic contexts hinted at problems with equating *H. sapiens* behavior solely with Upper Paleolithic archaeology (Binford 1968; Brose and Wolpoff 1970; Garrod 1962; Howell 1958). Yet it was not until the 1980s–1990s that archaeologists began to explicitly challenge the equation of Upper Paleolithic behavior with modern human behavior (White 1982). "Modern humans," "modern human behavior," and "behavioral modernity" rarely appear in the titles of books, articles in peer-reviewed journals, or dissertations before the 1980s (Minugh-Purvis 1995). These terms became common in the 1990s, when thermoluminescence, electron spin resonance, uranium series, optical stimulated luminescence, and radiopotassium dating showed that *H. sapiens* fossils previously thought to date to 40–50 kya or slightly older actually dated to 80–200 kya (Aitken, Stringer, and Mellars 1993). A series of prominent international conferences and resulting publications incorporating these terms into their titles vastly increased their popularity (Akazawa, Aoki, and Kimura 1992; Mellars 1990; Mellars and Stringer 1989; Smith and Spencer 1984; Trinkaus 1989). "Modern human behavior" and "behavioral modernity" allowed archaeologists to distinguish the recent part of the archaeological record that conforms to traditional European expectations about *H. sapiens* behavior from earlier phases of our species' evolutionary history that did not. Not all Paleolithic archaeologists employ this dichotomous framework. Gamble (1993:161), for example, distinguishes between "ancients" (Middle Paleolithic humans), "pioneers" (authors of Middle-Upper Paleolithic "Transitional" industries), and "moderns" (Upper Paleolithic humans); nevertheless, the underlying theoretical framework remains teleological. This idea of a significant evolutionary division between earlier *H. sapiens* and later "modern" ones, between us and an ancestral "them," has now spread well beyond Paleolithic archaeology into mainstream anthropology (e.g., Fuentes 2009:92).

Whining about Eurocentrism is fashionable in some academic quarters, but using one region, even a well-known one

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such as Europe, as a model for global patterns of human evolution inevitably risks equating the uniquely derived characteristics of human adaptation in that region with universal trends. If, for example, the first archaeologists had been Polynesians, the important hallmarks of behavioral modernity as they conceptualized it might include ocean-going watercraft, celestial navigation skills, pelagic fishing, hunting marine mammals, horticulture, domesticated pigs and dogs, ceramics, edge-ground stone axes, monumental architecture, and feather cloaks (Kirch 2002). Our Polynesian prehistorians would probably regard carved antler tools, cave art, and prismatic blade production as quaint local phenomena of no obvious evolutionary significance.

Narrative Traditions in Paleoanthropology

The notion that *H. sapiens* populations can be meaningfully divided into those who were behaviorally modern and those who were not fits well with narrative explanatory frameworks in paleoanthropology. As Landau (1991) demonstrated, many modern-day explanations of human origins ("anthropogenic narratives") retain the same structural elements as hero stories, folktales, and other mythological traditions (table 3). In all such anthropogenic narratives, the protagonist undergoes a fundamental change or "transformation" from an inferior earlier state to a subsequent superior one. This transformation is caused by a "donor." In prescientific narratives this donor was often a supernatural being or an object with magical powers. Early scientific accounts of human evolution identified such behavioral qualities as sociality, encephalization, terrestriality, and bipedalism as crucial donors in the early phases of human evolution. Much recent archaeological debate about modern human origins is a debate about the nature of this donor. Those previously discussed—neural mutations (Klein 2008), population pressure (McBrearty 2007), socioecological change (Ambrose and Lorenz 1990), and symbolic behavior (Henshilwood and Marean 2003; see also Mellars 2006*c*)—have been joined by age/sex-based division of labor (Kuhn and Stiner 2006), expanded social networks (Stiner and Kuhn 2006), and a broadened ecological niche (Marean 2007; O'Connell 2006; Shea 2007*a*). None of these phenomena are necessarily mutually exclusive, and thus much of the debate surrounding them concerns their relative importance for explaining particular archaeological evidence.

Table 3. Key elements of the anthropogenic narrative and corresponding events in recent models of *Homo sapiens* evolution

Order	Narrative function	Events in *Homo sapiens* evolution
1	Initial situation	Middle Pleistocene Africa
2	Introduction of the hero	Ancestors of *H. sapiens*
3	Change	Skeletal modernity
4	Departure	Adaptive radiation within Africa
5	Test/fail	Failure to disperse beyond Africa
6	Donor	Various
7	Transformation	Behavioral modernity/modern human behavior
8	Test again	Dispersal from Africa <50 kya
9	Success	Global settlement and ecological dominion

Source. Landau 1991.

Methodological Problems with Trait Lists

Narrative explanatory frameworks are satisfying ways of explaining complex phenomena, but their simple linear structure constrains the kinds of explanations we can propose about human origins. The most severe of these constraints is that narrative explanations require us to define "behavioral modernity" in terms of a list of discrete traits. Trait lists are a good starting point for defining anything, but there is a growing consensus that the trait lists most commonly used to define modern human behavior are critically flawed. Henshilwood and Marean (2003) identify three major flaws in the current use of trait lists to define "behavioral modernity": taphonomic bias, ambiguity, and a lack of theoretical justification.

Taphonomic bias. Some of the traits used to define "behavioral modernity," such as carving bone/antler tools, can only be recognized by the preservation of osseous tissues. Generally poor preservation of biotic remains from equatorial African and Asian contexts virtually guarantees (potentially false) negative findings about human behavioral modernity.

Ambiguity. Some of the subsistence-related "modern" behaviors are more plausibly explicable in terms of ecological factors, such as intensification, than they are in terms of some newly evolved capacity. Systematic collection of small game, fish, and birds entails high handling costs that make them a poor choice for small groups of human consumers. Thus, historical patterning in the occurrence of evidence for these behaviors may reflect variation in human population pressure rather than the evolution of some novel behavioral capacity.

Theoretical justification. It is not clear why many of the purportedly modern human behaviors are even included on trait lists. The use of seasonal mobility as a characteristic of modern human behavior suffers from a particularly weak theoretical justification. Not only do the mobility patterns of living humans vary widely with respect to this variable (Kelly 1995; Lieberman 1993; see references in Sellet, Greaves, and Yu 2006), nonhuman species plainly do so as well (e.g., wildebeest, reindeer, migrating birds).

Even if one could pare down the traits used to define "modern human behavior" to those that were minimally affected by preservation, were unambiguous, and were theoretically justifiable, the intrinsic variability of human behavior poses additional serious problems. These problems involve composition, relative significance, statistical significance, the limits of extrapolation, and linkage.

Composition. Most trait lists of modern human behavior

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contain some odd omissions. The use of long-distance projectile weaponry is universal among living and historical humans and is thus far associated solely with *H. sapiens* in the paleoanthropological record (Shea 2006*b*). All humans either control or manipulate the reproductive strategies of plants and/or other animals. All humans who have lived near substantial bodies of water make and use watercraft. These behaviors are no less complex, universal, or potentially significant for human dispersal and adaptive success than behaviors included on these trait lists.

Relative importance. Are all the behaviors on the list of equal diagnostic value, or are some more important than others? There is a growing consensus that exosomatic symbolic behavior is clear and conclusive evidence for behavioral modernity. Yet this consensus has at least as much to do with the ease of eliminating other explanations for such phenomena as it does with confident demonstrations that symbolic behavior was an important aspect of our species' evolution. The Middle Paleolithic *H. sapiens* populations whose remains have been recovered from Skhul and Qafzeh Caves in Israel in contexts dating to 80–130 kya practiced mortuary rituals (including burial with grave goods), color symbolism, and the use of personal adornments (Shea and Bar-Yosef 2005). However, these humans vanished from the Levant between 45 and 75 kya and were replaced in the local fossil record by Neanderthals, who also practiced mortuary rituals and who also became extinct locally after 45 kya (Shea 2007*a*, 2008*b*). The capacity for exosomatic symbolic behavior does not appear to have done either these Levantine early humans or Neanderthals much good in the long run.

Statistical significance. Claims that particular groups of *H. sapiens* and sometimes Neanderthals were behaviorally modern are often supported by tabulations of evidence for modern human behavior gathered from archaeological contexts dispersed over whole continents and spanning tens or even hundreds of thousands of years. This approach to recognizing behavioral modernity ignores the basic statistical principle that larger samples always show greater variability than smaller ones (unless, of course, they are affected by systemic bias). Thus, claims about human behavioral modernity are far more likely to be empirically justifiable for larger regions and longer periods (e.g., all of Africa between 40 and 250 kya) than for smaller and more circumscribed ones (say, Eritrea between 100 and 130 kya). There is currently no consensus among Paleolithic archaeologists about the appropriate geographic and chronological scope for such compendia of evidence. Nor is there a consensus about the proportion of contexts from a given region and time period at which affirmative evidence for a behavior is apparent for it to be accepted as having been credibly established. Even so seemingly reasonable an expectation of a simple majority (51%) encounters the problem that the archaeological record is not a probabilistic sample. With few exceptions, archaeological sites are clustered around modern-day transportation routes and population centers. They are in no sense probabilistic samples. That one can only use probabilistic samples to legitimately infer parametric variation from the statistical variability is a foundational principle of statistical inference (Sokal and Rohlf 1995).

Limits of extrapolation. How broadly can one extrapolate a capacity for a particular behavior from a small number of archaeological occurrences? How much earlier or later in time? Over how broad a geographic range? Would finding conclusive evidence for behavioral modernity in, for example, Kenya after 60 kya indicate similar capacities throughout East Africa? Throughout all of Africa? Throughout *H. sapiens*' geographic range at that point in time? Among hominin species thought to be immediately ancestral to *H. sapiens*, and if so, at what level of probability?

Linkage. By grouping behaviors together on a trait list, one is implicitly accepting the hypothesis that some kind of ultimate causal linkage exists among them. This might be plausible for some behaviors but not for others. For example, tasks that require the conjunction of several disparate components (e.g., hafted stone tools, polychrome mural art) might be connected by some shared cognitive capacity, but we do not actually know this. Nor do we know that these behaviors necessarily derive from the same structural underpinnings as the production of prismatic blades, fishing, carving bone points, or other putatively modern behaviors.

There is no archaeological consensus about the answers to any of these questions about the use of trait lists to define modern human behavior, and yet what more fundamental questions are there than how a scientific model deals with observational variability? The use of narrative explanations requires particular answers with respect to composition, relative significance, statistical significance, the limits of extrapolation, and linkage. "Modern human behavior" is those important and admirable elements of our behavior that we can credibly project back to some Pleistocene "Golden Age" (Roebroeks et al. 2001). Observations of particular behaviors can be extrapolated to the whole of humanity at any one point in time and from the first appearance of that evidence thereafter. All derived behaviors reflect the transformation caused by a particular evolutionary "donor."

What Is the Opposite of Behavioral Modernity?

Narrative explanation is a valuable tool. It would not have persisted and become so widespread cross-culturally for at least the last five millennia if it were not. Unfortunately, it is a tool that leaves us ill equipped to answer questions about the evolution of human behavioral variability. There are few clearer illustrations of how treating behavioral modernity as if it were a real thing constrains our ability to think about behavioral variability than to consider its alternatives. Just what is the opposite of behavioral modernity? Two terms in current usage are "nonmodern human behavior" and "archaic human behavior." "Nonmodern human behavior" reduces all hominin behavioral variability to a dichotomy, the least-

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detailed way of organizing any observations. "Archaic human behavior" arrays all of human behavior along a single continuum from "archaic" to "modern." Ethnologists discarded unilinear "evolutionary" models of human behavioral variability in the early twentieth century. There is no reason for Paleolithic archaeologists to continue to harness their theories to such outdated and simplistic models of human behavioral evolution and variability.

Behavioral Modernity versus Behavioral Variability

One can marshal countless principled arguments against behavioral modernity as an analytical construct, but the most convincing challenge to behavioral modernity is a refutation of its core hypothesis. Nearly all current models of modern human origins share the conviction that there were significant behavioral differences between the earliest *Homo sapiens* and those extant since 40–50 kya. This hypothesis can be tested with lithic evidence from the earliest archaeological sites associated with *H. sapiens* fossils.

Materials

The oldest-dated *H. sapiens* fossils come from archaeological sites in the northern part of East Africa (Eritrea, Ethiopia, and Kenya) dating to between 100 and 300 kya. The best-dated and best-documented of these sites are in Ethiopia and include the Middle Awash Valley Aduma and Bouri formations (Yellen et al. 2005) and the Herto Member of the Bouri Formation (Clark et al. 2003), Omo Kibish Members 1–3 (Fleagle et al. 2008; Shea 2008*a*), and Gademotta/Kulkuletti (Morgan and Renne 2008; Wendorf and Schild 1974; Wendorf et al. 1975). These are joined by sites from farther afield, the Abdur Reef Limestone Complex in Eritrea (Bruggemann et al. 2004), and the Kapthurin Formation of western Lake Baringo, Kenya (McBrearty 2005; Tryon 2006). Collectively, these sites include and bracket (both geographically and chronologically) the oldest-known fossil remains of *H. sapiens*. All of them, except Abdur, have been the focus of controlled archaeological excavations. This list could be significantly expanded by increasing its geographic scope or by including additional sites with less constrained geochronology and/or ones known mainly from surface collections (for a detailed review of this evidence, see Basell 2008), but doing so would introduce more than the minimum number of uncertainties into this comparison.

In testing hypotheses about behavioral variability, it makes sense to start with the most durable and ubiquitous archaeological evidence, in this case the lithic record. All these sites preserve stone artifacts (fossil preservation is more variable), but comparing the lithic evidence for human behavioral variability from these sites is not a simple proposition. The contexts vary in terms of lithic raw materials represented and their geological contexts, in the strategies used to recover stone tools (i.e., controlled area excavation, test trenching, surface collection), in the particular typologies used to describe the tools, and in the degree to which they have been published. Comparisons based on tool relative frequencies or statistical variation in measured properties of stone artifact variability are impossible owing to inadequate documentation, or even where possible, they have little meaning outside the context of individual sites because of differences enumerated above. This limits comparisons to nominal scale variability.

Methods

One popular framework for describing and comparing lithic variability was developed by Grahame Clark (1969:31). This framework describes lithic technological variability across the length of the Paleolithic in terms of five "modes" (table 4). Clark's modal framework is a crude instrument for measuring lithic variability, but it enjoys a broad measure of popularity among Paleolithic archaeologists in Europe, Africa, and elsewhere (Barham and Mitchell 2008; Foley and Lahr 2003; Gamble 1999). Most formulations of modern human behavior include regular use of mode 4 (prismatic blade) technology such as that seen in European Upper Paleolithic assemblages (but see Bar-Yosef and Kuhn 1999; Foley and Lahr 1997).

Table 4. Clark's lithic technological modes

Mode	Characteristics	Traditional Paleolithic terminology	Diagnostic criteria used in this study
1	Pebble cores and flake tools	Lower Paleolithic Oldowan	Choppers, discoids, polyhedrons
2	Large bifacial cutting tools made from flakes and cores	Lower Paleolithic Acheulian	Picks, handaxes, cleavers, lanceolate bifaces
3	Flake tools struck from prepared cores	Middle Paleolithic, Middle Stone Age	Levallois cores, flakes, points, asymmetrical discoidal cores
4	Punch-struck prismatic blades retouched into various specialized forms	Upper Paleolithic	Prismatic blade cores and blades
5	Backed microliths and other retouched components of composite tools	Mesolithic, Epipaleolithic, Later Stone Age	Backed microliths

Source. Clark 1969:31.

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Mode 5 technology, the production and use of geometric microliths, is generally seen as a later Upper Paleolithic or Postglacial phenomenon in Europe.

One problem with using Clark's modal framework to assess technological variability among East African later Middle Pleistocene assemblages is that it treats the different technological modes as an evolutionary progression.[1] That is, it emphasizes the most derived features of an assemblage while remaining mute about the retention of archaic/primitive components. This is problematic because it underestimates variability. For example, an assemblage with 44 pebble cores and five handaxes would be classified as mode 2 even though the latter large cutting tools (characteristic of mode 2) were vastly outnumbered by core tools characteristic of mode 1. Adding a handful of blades and a blade core or two would transform it into a mode 4 assemblage and fail to register the absence of the prepared cores and flake tools characteristic of mode 3 assemblages. To preserve a register of variability in characterizing East African Later Middle Pleistocene assemblages using this framework, table 5 notes the diagnostic evidence for all five of Clark's lithic technological modes. If an assemblage meets the criteria for more than one of these modes, this is indicated in table 5.

Table 5 also presents lithic data from several reasonably well-dated and well-described Late Pleistocene and Early Holocene archaeological contexts from the same general region as the Later Middle Pleistocene sample. The sites in question include Porc Epic (Ethiopia; Pleurdeau 2005), Lothagam and Lowasera (Kenya; Phillipson 1977; Robbins 1974), and Lukenya Hill 2 (GvJm16 and GvJm22; Barut Kusimba 2001).

Results

The archaeological assemblages found with the oldest *H. sapiens* in East Africa and those from roughly contemporary sites preserve evidence for mode 1–4 lithic technologies. This amounts to four-fifths (80%) of the range of lithic technological modes with which *H. sapiens* is associated throughout Africa and most of Eurasia after 50 kya. At least insofar as stone tool production is concerned, the earliest *H. sapiens* in East Africa were capable of as great a range of behavioral variability as humans associated with mode 4 assemblages but not mode 5 ones, that is, Upper Paleolithic Europeans. There are differences among the range of lithic technological modes represented in individual samples, but few recent archaeological sites feature evidence for the full range of modern human behavior either.

Stone tool assemblages made by Late Pleistocene and Holocene Africans who lived in the same region as the earliest *H. sapiens* differ from these earlier ones mainly in preserving evidence for microlithic mode 5 or microlithic technology. Microlithic technologies occur in African contexts sporadically between 50 and 100 kya (Ambrose 2002). They become common and widespread after 20 kya, not just in Africa but also in Eurasia (Elston and Kuhn 2002). However, one ought not read too much into this difference. Small retouched stone tools are known from many Eurasian Lower and Middle Paleolithic contexts (Dibble and McPherron 2006; Zaidner, Ronen, and Burdukiewicz 2003). Small geometric-backed pieces are known from Last Interglacial (presumably Neanderthal) contexts in Germany (Conard 1990). Whatever cognitive capacities mode 5 microlithic technologies require were plainly ones that were either evolutionarily primitive or evolved convergently in the genus *Homo*.

These observations challenge the assumption that there were significant differences in the capacity for behavioral variability between the oldest-known *H. sapiens* and recent humans. In other words, the evidence is insufficient to reject the null hypothesis of "no difference" between the earliest *H. sapiens* and behaviorally modern Late Pleistocene ones in term of their measurable behavioral variability.

Discussion

Is this test of behavioral variability an adequate test of early *H. sapiens*' behavioral modernity? Stone tools are less vulnerable to preservation biases and taphonomic filters than other kinds of evidence, but one could argue that stone tool technology alone is too narrow a perspective from which to frame so broad a conclusion. Partisans of symbolic behavior as the defining characteristic of behavioral modernity could argue that the lithic record is unlikely to provide a clear and unambiguous test of their hypothesis. Such an objection is reasonable, and one could reply to it by introducing plausible supporting evidence for some exosomatic symbolic behavior. Recent excavations at Middle Stone Age sites in the Lower Omo Valley Kibish Formation preserve lithic evidence that could indicate a symbolic capacity similar to that asserted as proof of behavioral modernity in other African contexts. It is equivocal evidence (Shea 2008*a*) but no less equivocal than other claims for symbolic behavior in remote antiquity.

Among the lithic assemblages comprising the Kibish Industry, there is a wide range of lithic raw materials, including basalt, rhyolite, and a variety of cryptocrystalline silicates (chert, chalcedony, jasper, and fossilized wood). In all three of the excavated assemblages from Omo Kibish, artifacts made of red materials are more heavily reduced (cores and flakes smaller) than other raw materials (Shea 2008*a*:470). This preference is most clearly observable among red jasper, a raw material that is scarce in gravel deposits, but it is also expressed in red shale and basalt. Metric differences between these artifacts (core length, flake length) do not rise to the level of

1. Another problem, again a reflection of Clark basing these modes on the European Paleolithic succession, is that they do not assign a diagnostic role to the production of foliate bifacial points. These points are widespread throughout the later phases of the African Middle Stone Age and are seen by many Africanist prehistorians as evidence for the use of projectile technology (Shea 2006*b*).

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Table 5. Representation of technological modes for selected East African Later Middle Pleistocene contexts

Context	Date (kya)	Mode				
		1	2	3	4	5
West Baringo, Kapthurin Formation, Kenya	234–284	+	+	+	+	. . .
Gademotta/Kulkuletti, Ethiopia	175–278	+	+	+	+	. . .
Omo Kibish Members 1–3, Ethiopia	>104–195	+	+	+	+?	. . .
Middle Awash Valley, Herto Member, Bouri Formation, Ethiopia	160	+	+	+	+?	. . .
Abdur Reef Limestone Complex, Eritrea	125	+?	+	+	. . .	. . .
Middle Awash Valley, Aduma and Bouri MSA 1–3, Ethiopia	80–100	+	+	+	+?	. . .
Porc Epic Cave, Ethiopia	61–78	+	. . .	+	+?	+
Lukenya Hill 2 (GvJm22, GvJm16), Kenya	13–17	+	+	+	+	+
Lothagam, Kenya	6–7	+	+	+	+?	+
Lowasera, Kenya	<7	+	+	+	. . .	+

Note. Question marks (?) indicate identifications based on a relatively small number of specimens. For references, see text.

statistical significance. (This is probably because no measurements were made on the vast quantities of microdebitage [flake fragments shorter than 20 mm] recovered from these sites.) Even if such measurements indicated a statistically significant difference, they would not unambiguously indicate exosomatic symbolic behavior, because it is possible that red jasper has some desirable functional qualities that are not apparent to us. (During the excavation of the KHS locality, one Mursi crew member informed us that red jasper was particularly "good" for use in scarification when obsidian/glass or steel razors are not available.)

A second possible line of symbolic lithic evidence involves transport of nonutilitarian materials. At both KHS (the site from which the Omo 1 human was recovered) and AHS (the findspot of Omo 4), the lithic artifacts included small quantities of a greenish white opal silica (Shea 2008*a*:450). This rock is sharp, but it is also extremely brittle and does not fracture conchoidally in a predictable way. The nearest sources of this opal silica are a series of hot springs located some 15–20 km north of these sites. This evidence could be seen as proof of long-distance transport of nonutilitarian materials, and therefore potentially symbolic materials, but this is not necessarily clear and convincing evidence of symbolic behavior. The poor conchoidal fracture qualities of this opal silica were not apparent to me (an experienced flint knapper) until I had transported samples several tens of kilometers from the source to our field camp. It is not beyond the realm of possibility that early *H. sapiens* sometimes made similar bad decisions about raw material transport, decisions whose lithic by-products were no less durable than those resulting from wiser choices.

This does not exhaust the range of possible arguments that could be made from the Omo Kibish evidence or other later Middle Pleistocene sites in East Africa about early human behavioral modernity, but further such appeals serve little purpose. Behavioral modernity as it is currently conceptualized in human origins research is such that our ignorance about archaeological formation processes poses no obstacle to claiming any archaeological observation as evidence of modern human behavior.

Archaeologists do not yet know the underlying strategic calculus that governed the toolmaking behaviors indicated by the various technological modes, but we are getting closer to such knowledge. Following the lead of colleagues in primatology and human behavioral ecology (Tooby and DeVore 1987), archaeologists increasingly try to conceptualize and understand lithic variability through "strategic modeling" (Ambrose 2001; Kuhn 2004; Shea 2007*b*; Torrence 1989). Costs and benefits of transport and yield of cutting edge figure prominently in recent debates about Oldowan/mode 1 technologies (Braun et al. 2005; Kimura 1999, 2002). Debates about the distinctive large cutting tools that mark mode 2 technologies now focus on questions of transport efficiency and compromises between service as a core and as functional cutting tools in their own right (Kelly 1988; Toth 1997). Similar kinds of discussions are becoming increasingly common concerning Levallois prepared-core techniques, prismatic blade technology, and microlithization (Elston and Brantingham 2002; Eren, Greenspan, and Sampson 2008; Sandgathe 2004; Torrence 2002). Archaeological lithic analysis has not yet reached the point where we can say why later Middle Pleistocene East African humans made and discarded handaxes in some contexts and prismatic blades in others (or why none of them made geometric microliths). Nevertheless, what little we do know about the sources of variation in lithic technology from ethnographic and experimental sources is sufficient to support the hypothesis that those choices arose from variability in human behavioral strategies.

Uniformitarian theory compels us to seek the sources of human behavioral variability in factors we see influencing the behavioral strategies of recent humans. The most obvious of these factors include energetic costs, benefits, risks, and anticipated fitness consequences. The values of these variables can vary widely and nondirectionally at many different chronological and geographic scales and in response to situational variables. Perhaps it is not the particular modalities of these

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behavioral strategies that distinguish *H. sapiens* from other hominins but rather the quality of their variability. Maybe it is not whether we make blades or bone harpoons but how rapidly we switch from one strategic modality to another or how many and what kinds of strategies remain "in play" in a given ecological context that is unique to *H. sapiens*. Could there be specific combinations or cost, benefit, and risk that *H. sapiens* alone can sustain? These kinds of questions can only be answered by viewing the archaeological record through the lens of strategic behavioral variability. Forcing behavioral variability into a simple progression from a conjectural "archaic" ancestral state to "behavioral modernity" adds a layer of interpretation that neither clarifies our archaeological observations nor improves inferences about the course of human evolution.

Strategic Perspectives on Human Behavioral Variability

Most of the problems attending the terms "modern human behavior" and "behavioral modernity" can be eliminated simply by abandoning the use of these terms (Shea 2007*a*). Doing so forces one to describe particular behaviors in terms of the behavior itself, its component strategic variables, and its archaeologically detectable consequences. Discarding the concepts of modern human behavior and behavioral modernity shifts the focus of research on the evolution of *Homo sapiens* behavior away from unanswerable (indeed oxymoronic) questions about the derivation and evolutionary polarity of behaviors and closer to testable hypotheses about strategic sources of variability in particular human behaviors.

The Problem-Solving Value of a Strategic Perspective

Thinking about human behavior in strategic terms enables us to explain previously inexplicable phenomena in the archaeological record of recent human evolution. These problems include behaviorally modern Neanderthals, recursive behavioral modernity in *H. sapiens*' African archaeological record, and the elusive character of evidence for modern human behavior in Late Pleistocene and Early Holocene Sahul (Australia and New Guinea).

Behaviorally modern Neanderthals. Some of the particular lines of evidence cited as proof of African Middle Stone Age humans' and Eurasian Upper Paleolithic humans' behavioral modernity also occur in European contexts associated with Neanderthals (d'Errico 2003; d'Errico et al. 2003; Zilhão 2007). Much of this evidence is concentrated in Europe during the millennia immediately preceding Neanderthals' last appearances in the fossil record, a time when *H. sapiens* populations were dispersing into Europe (Mellars 2006*b*). Thus, there is some question of whether such modern behavior is an entirely independent development among Neanderthals (Mellars 2005). Allowing that it was would require us to reconsider much of what we think we know about Neanderthal–*H. sapiens* evolutionary relationships. Neanderthals are, for better or worse, the quintessential evolutionary "other." Historically, how we have thought about them has influenced how we think about "race" and other issues surrounding contemporary human variation (Trinkaus and Shipman 1993; Zilhão 2006*b*).

A strategic perspective parsimoniously explains similarities between Neanderthals and *H. sapiens*' archaeological records in terms of convergence, a well-known evolutionary phenomenon. Similar selective pressures can elicit similar strategic solutions among morphologically similar organisms (Grant 1972). Neanderthals and *H. sapiens* were hominin species with a recent last common ancestor. They lived at more or less the same time in similar habitats and with similar needs for food, shelter, and tools. In some parts of Europe, they lived in exactly the same environments within a few thousand years of one another (Blockley, Bronk Ramsey, and Higham 2008; Finlayson and Carrión 2007; Jöris and Adler 2008; Jöris and Street 2008). It would be astonishing if there were not superficial similarities in their archaeological records arising from evolutionary convergence in some of their archaeologically visible adaptive strategies. Europe is not unique in this respect; the much longer (45–130 kya) Middle Paleolithic archaeological record of the East Mediterranean Levant preserves abundant evidence for behavioral convergence between Neanderthals and *H. sapiens* in settlement patterns, technology, and subsistence (Shea 2006*a*).

Recursive behavioral modernity in Africa. If behavioral modernity was both a derived evolutionary condition and a landmark development in the course of human evolution, one would hardly expect it to disappear for prolonged periods in our species' evolutionary history (Hovers and Belfer-Cohen 2006). The best-known example of such "recursive" behavior involves the South African Middle Stone Age Howiesons Poort Industry, dated to 54–70 kya (Lombard 2005). Howiesons Poort stone tool assemblages feature prismatic blades and backed pieces, hafting, systematic use of exotic and/or rare local high-quality raw materials, the use of ochre, and bone tools. Many researchers have been struck by the "modern" behavior implied by the Howiesons Poort evidence. Indeed, prismatic blades and backed pieces made from blades occur all over Africa and Eurasia after 20 kya, plainly indicating that these toolmaking techniques were useful in a wide range of contexts. Yet, surprisingly, these distinctive "modern" behaviors are absent from immediately post–Howiesons Poort contexts in South Africa, and they do not appear to have spread widely throughout sub-Saharan Africa as a "package." The distinctive behavioral components of the Howiesons Poort industry do not appear again consistently in this part of Africa until well into Later Stone Age times.

Viewing the recursion of behavioral modernity implied by

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the Howiesons Poort evidence in strategic terms does not require one to explain the loss of behavioral modernity. The particular behaviors whose residues define the Howiesons Poort Industry each had costs and benefits. Prismatic blade production, for example, requires high-quality rocks and a certain considerable (if imprecisely quantifiable) level of skill (Eren, Greenspan, and Sampson 2008). It yields rectangular fracture products that are consistent in size and shape and thus readily inserted into previously shaped wooden handles, an arrangement that minimizes the costs of transporting lithic raw materials. A reduction in residential mobility might decrease incentives for knapping such readily haftable tools. The otherwise inexplicable failure of distinctive patterns of Howiesons Poort stone tool production to disperse northward into equatorial Africa could simply reflect relatively low residential mobility among humans living in equatorial African woodlands and a correspondingly lowered demand for hafted stone tools. Whether or not this particular hypothesis is sustained by future research, the important point is that a strategic perspective offers a credible and parsimonious explanation for recursive patterns of modern human behavior. Such recursions are not readily explicable by any model in which behavioral modernity is a real, derived, and essential biologically based quality.

Elusive modern human behavior in Sahul. New Guinea and Australia were demonstrably occupied by *H. sapiens* from 40 to 45 kya onward, and yet few of the traits used to define modern human behavior appear in this region consistently until mid-Holocene times (Habgood and Franklin 2008; O'Connell and Allen 2007; Stern 2009). It is difficult to reconcile this evidence with the notion of behavioral modernity as a significant and universal human quality. There is no evidence for any Pleistocene "human revolution" in Sahul comparable with that observed in western Eurasia. Indeed, on Tasmania, carved bone tools and the systematic use of marine resources were developed in Pleistocene times and lost again in the millennia immediately preceding the arrival of Europeans. Taking this evidence at face value would imply that behavioral modernity was lost, not once but twice. The course of behavioral modernity in Tasmania looks a lot more like behavioral variability than it does a "human revolution."

The notion that Late Pleistocene Australasians were deficient in their behavioral modernity is based mainly on the relative simplicity of their stone tool technology. The stone tool assemblages found at the earliest Australian sites are grouped together and described as part of a "Core-Tool and Scraper Tradition" (Mulvaney and Kamminga 1999). These lithic assemblages differ little from Lower Paleolithic Eurasian ones in most key respects, and they would almost certainly be referred to mode 1 in Clark's (1969) framework. If one thinks about behavior as a product of strategies, differences in historical patterns of human behavioral variability in Eurasia and Australasia are easily explicable. The two continents are separated from one another by a major biogeographic frontier (Wallace's Line), and they differ profoundly from one another in ecological terms. It is hardly surprising that stone tool technology in Sahul differs from that deployed at the same time at the opposite end of the Eurasian continent.

Viewed in strategic terms, mode 1 technology is the most versatile stone tool technology ever devised by hominins (Shea 2007*b*). Occurrences of mode 1 assemblages range from the Late Pliocene to the ethnographic present and from Equatorial Africa to the Arctic. That early Australasian *H. sapiens* chose to manufacture mode 1 stone tools plausibly reflects a rational and intelligent choice of a simple, versatile, and thus evolutionarily stable lithic component for their technological strategies. Professional flint knappers excepted, smart people do not set out to maximize the amount of time they spend knapping razor-sharp stone tools with their bare hands. As the Aranda (Aboriginal Australian) proverb says, "The more you know, the less you need."

The appearance of microlithic tools, edge-ground tools, and other more complex stone technology in Australian sites after 5–6 kya coincides closely with the establishment of modern sea levels—no trivial thing for a continent with a relatively broad and shallow continental shelf (Lourandos 1997). This late appearance of more "modern" ways of knapping stone is plausibly related to socioecological changes associated with the establishment of modern coastal environments. Recent human coastal adaptations in Australia and elsewhere are marked by relatively high degrees of technological complexity and intensified subsistence practices (Erlandson 2001). At present, it is not known whether this late appearance of modern human behavior reflects novel phenomena associated with a stable coastline or if it simply marks the first point in time when coastal adaptations formerly practiced on now-flooded landscapes are preserved in sedimentary deposits accessible to archaeologists. Either one of these hypotheses is a vastly more plausible explanation than one invoking the "loss" and recent "re-evolution" of behavioral modernity among Aboriginal Australasians.

Why Discard Behavioral Modernity as an Analytical Construct?

The vast number of journal articles, books, conferences, documentaries, and press releases incorporating "modern human behavior," "human behavioral modernity," and allied terms show Paleolithic archaeologists and others have found these concepts useful. Arguments could be made for retaining these concepts in human origins research, but the arguments against doing so are far more compelling.

The strongest reason for discarding "behavioral modernity" and "modern human behavior" is that they lack analytical precision. As matters stand today, there are wide and irresoluble theoretical disagreements about the nature of behavioral modernity, how to define it, and how to recognize it. Eurasian prehistorians use the term "modern human behav-

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ior" for evidence that occurs consistently over tens of thousands of years at a regional scale (various parts of Europe and/or Southwest Asia). Africanists use the term for behavior that occurs intermittently over hundreds of thousands of years at a continental scale. Neither term clarifies the description of archaeological evidence, nor does either of them refine our understanding of the evolution and variability of a particular behavior. They have become postmodern concepts, words that mean whatever one wants them to.

The idea that behavioral modernity is a derived evolutionary state, one not shared by all morphologically modern-looking *H. sapiens* and one that can be reliably diagnosed from behavioral characteristics, is rich with potential for abuse. It fits well with racist arguments that there are meaningful grade-level evolutionary differences among living humans. Such views are rarely expressed in scientific circles (or polite company), but they nevertheless can find traction among nonscientific audiences because they incorporate the same unilinear model of human evolution that underlies the behavioral modernity concept. If paleoanthropologists judge humans' evolutionary state based on their behavior, why shouldn't others do so as well? Discarding the term "behavioral modernity" will not stop individuals from cherry-picking selected findings of paleoanthropology to support racist agendas, but it will deny them the illusion that they are emulating an accepted scientific method.

Creationists do not mock paleoanthropologists for using the terms "modern human behavior" or "behavioral modernity," but they would be justified in doing so. These terms imply that our evolutionary history is teleological, that it follows a trend or predetermined trajectory. As with the evolution of all living things, human evolution was a complex and contingent process. Long-term trends are perceptible only in retrospect (Gould 1987). (This is why long-term predictive modeling of human evolution is filed in the science fiction sections of most bookstores.) Refocusing archaeological research on the causes of historical patterns in human behavioral variability will remove a potential source of embarrassment for paleoanthropology. Creationists will still cite the older literature that refers to behavioral modernity, but abandoning this term will deprive them of the pretence that they are addressing contemporary issues in human origins research.

Modern human origins has been the most hotly debated topic in paleoanthropology for the last three decades. It has particularly energized archaeological research on the African Middle Stone Age and the Middle-Upper Paleolithic Transition in western Eurasia as well as historical linguistics and molecular anthropology. Consequently, there are strong professional or "careerist" incentives to retain behavioral modernity as an analytical construct. Describing research as germane to "modern human origins" implicitly stakes a claim for priority in attention, publicity, and funding. Discarding "behavioral modernity" will not stop individuals from hitching their wagon to a hot topic because this strategy reflects underlying selective pressures in contemporary academia (White 2000). But it will create selective pressure for individuals to explain the relevance and significance of that research on its own terms. Archaeological research that is genuinely, crucially, and centrally relevant to the evolution of *H. sapiens* does not need to engage in deceptive adverting. Its merits speak for themselves.

This article is not calling for any sort of ban on the terms "behavioral modernity" or "modern human behavior." Science does not need speech codes. In hypothesis testing and peer review it already possesses ruthlessly effective mechanisms for disposing of wrong or useless ideas. Rather, it calls on Paleolithic archaeologists researching the evolution of *H. sapiens* to conceptualize human behavior in strategic terms, to seek the cost-benefit structure of the incentives underlying particular behaviors, and to document variation in the contexts in which particular behavioral strategies are deployed (or not). The result will be far better models of human behavioral variability than are currently available. In doing this work, archaeologists will of necessity have to work more closely with behavioral ecologists. This field has developed a sophisticated and nuanced language for describing and analyzing strategic and behavioral variability of living humans and other species. Such collaboration could be a very good thing for archaeology and for anthropology in general. A common focus on strategic modeling of human behavior is a powerful antidote to the centrifugal forces pulling anthropologists away from one another into ever more rarified specializations.

Conclusion

Using evidence from stone tool variation, this article has shown that the earliest *Homo sapiens* populations who lived in East Africa around 200 kya possessed a capacity for behavioral variability identical to "behaviorally modern" Upper Paleolithic humans. The long-standing assumption that there are vast behavioral differences between these earliest humans and so-called modern humans like ourselves is almost certainly wrong. How much farther back in time this capacity for behavioral variability extends remains unknown. This conclusion has three important implications for Paleolithic archaeological research on the origins and evolution of *H. sapiens*. First, the capacity for behavioral variability we think to be uniquely evolved among recent human populations may be evolutionarily primitive. Second, this capacity for behavioral variability may be one shared with now-extinct hominin species. Finally, differences in the capacity for behavioral variability may not explain why these other species are extinct and *H. sapiens* is not. The case for behavioral variability is a strong one, but few major issues in evolution boil down to single causes.

Understanding why *H. sapiens* survived and other species did not requires us to make fundamental changes in the ways

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Paleolithic archaeologists investigate prehistoric hominin behavior. Paleolithic prehistory is essential to an anthropological understanding of human uniqueness and variability. More than 90% of the time that morphologically modern-looking *H. sapiens* has been in existence falls within the Paleolithic period. Archaeological sites vastly outnumber hominin fossils for all but the earliest (late Pliocene) phases of Stone Age prehistory. Yet archaeologists' potential contribution to anthropological debates about human uniqueness is severely limited by our habit of using "modern human behavior" and "behavioral modernity" to place human behavior in an evolutionary framework. In this article I have argued that there are no such things as modern human behavior and behavioral modernity. There are merely patterns of behavioral variability associated with *H. sapiens* in different archaeological contexts. Detecting which behaviors and qualities of behavioral variability are derived solely among *H. sapiens* requires archaeologists to examine the strategic sources of variability in the Paleolithic record.

Behavioral differences between "behaviorally modern" humans and their precursors are framed dichotomously and are only explicable in tautological terms. Early *H. sapiens* did not behave like modern humans because they lacked some ineffable quality of "behaviorally modernity." Humans living 200,000 years ago on the banks of the Omo River Valley behaved differently from the Nyangatom people who live there today, differently from Upper Paleolithic Europeans, and differently from modern-day college professors. The really interesting questions about the differences among these humans are not that differences exist but, rather, why particular differences exist. (This is, after all, anthropology's second self-defining question.) What are the underlying sources of behavioral variation and their costs, benefits, and long-term evolutionary consequences? Trying to "evolve" the Omo Kibish humans into us by invoking behavioral modernity does nothing but dredge the mythological precursors to paleoanthropology for explanations of human origins. The earliest *H. sapiens* were no less complete versions of us than we are incomplete versions of whatever our species will look like 200,000 years from now. We are merely different from one another.

That we are defined by what we do and not vice versa is an admittedly American sentiment, but it is one appropriate to research on *H. sapiens* evolution. We do not use behavioral characteristics to define *H. sapiens*, but we do frame our answers to broader philosophical questions about what it means to be human in terms of the behavioral variability associated with *H. sapiens* in the fossil and archaeological records. This article proposes that Paleolithic archaeology abandon the fruitless and divisive search for "modern human behavior" and instead refocus our energies on understanding the behavior of *H. sapiens* in all its variability.

The search for interpretive principles, or "middle-range theory," in Paleolithic archaeology most often involves a search in the ethnographic record of recent human hunter-gatherers. This is appropriate from a uniformitarian epistemological standpoint, but it is also unnecessarily self-limiting. The behavioral strategies practiced by recent human hunter-gatherers have variable time depth (Marlowe 2005). The particular combinations of strategies Pleistocene humans deployed in any particular context may have differed widely from those of recent human foragers. Whether one is interested in hunting technology (Churchill 1993), aquatic resources (Erlandson 2001), the use of personal adornments (Kuhn and Stiner 2007), or some other behavior, there is no reason to restrict one's sources of information to the world's dwindling number of hunter-gatherer societies. We must remain alert to the possibility that the behavioral strategies of earlier hominins may differ markedly from the behavior of recent humans (Lieberman et al. 2007). Sources for strategic models of prehistoric human behavior are not limited to hunter-gatherers, to humans, or even to primates (Tooby and DeVore 1987).

Reorganizing our search for human uniqueness into research on the strategic sources of human behavioral variability decouples Paleolithic archaeological research from a narrative explanatory framework that severely constrains the kinds of hypotheses we can propose and test about behavioral change in *H. sapiens* evolution and other hominins. This is not a new agenda for Paleolithic research but instead a call to return to a very old epistemological one. In his *Meditations*, Marcus Aurelius (AD 121–180) wrote "Of each particular thing ask: What is it in itself? What is its nature?" We need to heed his advice with respect to modern human behavior and behavioral modernity. Calling a particular behavior "modern" clarifies neither the occurrence of evidence for that behavior in a given context nor the sources of its variability. It merely adds a layer of interpretation between ourselves and observations of a particular phenomenon. By looking at the strategic underpinnings of particular behaviors, the costs of prismatic blade production, the benefits of making carved bone tools, the risks of long-distance exchange networks, the consequences of a broad ecological niche, and similar such questions, Paleolithic archaeology will make a substantive contribution to the core questions about human uniqueness and behavioral variability that define anthropology.

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Comments

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In paleoanthropology, as in other disciplines, particular concepts or models occasionally become entrenched to such an extent that their actual validity is rarely if ever questioned. However, when such appraisals are undertaken, the results can often lead to a revolution in the way research is essentially structured and conducted (Kuhn 1962). For several decades Paleolithic archaeologists have spent considerable time and resources investigating the origin of modern human behavior and the uniquely derived behavioral characteristics that distinguish *Homo sapiens* from their immediate evolutionary predecessors and other "archaic" hominins. This has been accomplished largely through appeals to the concept of behavioral modernity, but as Shea argues, it is now time for a fundamental shift in the way we conceive of hominin behavior and how Paleolithic archaeologists attempt to examine it.

Shea challenges the primacy and efficacy of the modern-archaic behavioral dichotomy in paleoanthropology and behavioral modernity in particular, which he attacks primarily for its qualitative and essentialist qualities and which he argues is inadequate at all epistemological levels. Following a thorough rebuttal of the behavioral modernity concept and its application by Paleolithic archaeologists, Shea makes a detailed case for the study of human behavioral variability. He stresses the need to think about human behavior strategically, in terms of costs and benefits, as a way to develop testable hypotheses about the behavior of *H. sapiens* that is theoretically rigorous, statistically valid, and conceptually independent from traditional, typically erroneous notions of behavioral evolution.

I applaud Shea for his explicit and wide-ranging critique of behavioral modernity as an analytical construct, the methods he employs to demonstrate the acute limitations of the concept, and the productive alternative he proposes. In point of fact, a small but increasing number of Paleolithic archaeologists from around the world is pursuing a research agenda similar to that espoused by Shea, concentrating on issues of behavioral variability as frequently inferred through lithic technology, foraging economy, and mobility, with reference to the human behavioral ecology literature (e.g., Adler et al. 2006; Braun et al. 2008; Elston and Brantingham 2002; Kuhn 2004; Kuhn and Stiner 2006; McCall 2007; Stiner and Munro 2002; Wallace and Shea 2006). However, because the vast majority of Paleolithic archaeologists and others outside the discipline (e.g., Hill, Barton, and Hurtado 2009) remain focused on behavioral modernity and its allied concept, modern human behavior, and the quest for its origins, Shea's polemic is both timely and germane.

Shea's arguments are of equivalent value to archaeologists working outside Africa, in particular Eurasia, where debates concerning Neanderthal biological and cultural competence have by tradition hinged on direct comparisons with behaviorally modern Upper Paleolithic humans and the trait lists used to define "behavioral modernity." In essence, Neanderthals have been largely defined by what they were not (i.e., modern humans) rather than what they were—a geographically widespread, culturally sophisticated, behaviorally variable, highly adaptable, and long-lived hominin. As more Paleolithic archaeologists working in Eurasia distance themselves from such restrictive comparisons and begin to focus their attention on behavioral variability, assessing the cost and benefits associated with specific behaviors without reference to other hominins, the vast behavioral and cognitive gulf that was once thought to exist between the Neanderthals and modern humans will continue to narrow.

I am in full support of Shea's reasoned critique and his proposed alternative; thus, the few faults I do find with his present efforts are of limited import. I believe Shea's call for the abandonment of the behavioral modernity concept, while certain to engender lively debate within particular academic circles, will succeed as research on the "behavioral variability" of hominins continues to yield significant results that challenge our traditional models of past human behavior and ultimately what it means to be human.

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Shea adds a significant voice to the growing chorus for the abandonment of behavioral modernity as an analytical concept. He neatly outlines the inherent (and inherited) flaws in the concept and rightly urges us to devote scarce time and resources to the more productive goal of elucidating the evolution of behavioral variability rather than seeking illusory markers of modernity. Others have made similar calls, arguing that behavioral ecology offers the most effective framework for examining behavioral variability at differing temporal and spatial scales (e.g., Barham 2007; Deacon and Wurz 2001; Foley and Gamble 2009; Gamble 2003; James and Petraglia 2005; Marean 2007; McBrearty 2007; O'Connell 2006; Potts 2001; Zilhão 2007). There is even a congruence in titles between Shea's "*Homo sapiens* Is as *Homo sapiens* Was" and "Modern Is as Modern Does" (Barham 2007).

What differentiates Shea's contribution from most others is his explicit challenge to the widespread assumption that early *Homo sapiens* lacked the behavioral flexibility exhibited by later humans. I agree, but let us extend that challenge to include other later Middle Pleistocene hominins: why retain

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the anatomical Rubicon of *H. sapiens* when dealing with behaviorally variable species such as *Homo heidelbergensis*?

Most archaeologists still argue that early *H. sapiens*, let alone *pre-sapiens*, provides little material evidence for having symbolically structured societies and as such is not recognizably modern (e.g., Henshilwood and Marean 2003). This may indeed be the case if we equate behavioral modernity with markers of external symbolic storage (art, artefact style, socially structured use of space), but as Shea observes from the perspective of the ethnography of East Africa, much contemporary expression of art and style takes place on perishable materials that would not survive in the archaeological record. Taphonomic filters have no doubt played a role in biasing our perception of the symbolic capacities of earlier hominins, but we work with what we have and build our models on existing data rather than on speculating on what might have been (Chase and Dibble 1987).

I do not wish to prolong the debate over absence of evidence versus evidence of absence of symbol use, but we must also be cognizant of the epistemological dilemma faced when considering evidence for early symbol use. As G. A. Clark (2002) observed, that evidence may take forms that we simply do not recognize given our existing expectations of what constitutes symbol use. We live in a symbol-saturated world, and to see alternative expressions of this capacity may be difficult given our current reliance on unambiguous signals. Other indirect markers of symbolically structured social systems exist today, as in the case of complex technologies that involve considerable investment in social learning. Perhaps we need to think laterally to find archaeological correlates (Barham 2010).

Shea offers some brief guidance here. As well as taphonomic limitations obscuring our vision of past capabilities, there may have been times when there were "weak incentives" or strong disincentives for investing in durable symbols. These were presumably a mixture of demographic, social, historical, and ecological forces, and here behavioral ecology offers testable predictions that can be applied archaeologically (O'Connell 2006). Powell, Shennan, and Thomas (2009:1301) have modeled the "critical effective population size, and therefore density, necessary for the accumulation of markers of modern behaviour." Modernity in this sense is not species specific but a product of the coalescence of certain pressures in certain places at certain times, places and times that can be estimated. This formulation should in theory be applicable to other hominins, such as Neanderthals (e.g., Zilhão et al. 2010), which had the capacity to transmit culturally inherited skills. The logical extension of breaking the linkage between species and behavior is the recognition that external symbolic storage is an emergent rather than uniquely derived property of Late Pleistocene *H. sapiens*. The label "demographic modernity" captures this phenomenon of a contextually contingent florescence of symbolic behaviors (Barham, forthcoming) and has obvious implications for understanding the recursions and convergences in behavior highlighted by Shea.

Shea also challenges us to reconsider the taphonomic and methodological biases that can unwittingly structure the archaeological record and color our interpretations of process between the extremes of gradualist and punctuated change. Such a reminder is timely if a new paradigm is to emerge from the loosening of the intellectual shackles imposed by the concept of modernity. We will be better placed to shift our gaze toward long-term trends, away from the distorting effect of first appearances (e.g., Gowlett 2009), and at the same time focus on the complex interplay of variables that underpin particular localized behaviors. A clearer understanding of the evolution of behavioral variability will result. The more influential voices that join this chorus, the more likely we are to sing in harmony.

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6 VIII 10

Because we know that as recently as a few tens of thousands of years ago, at least four forms of hominins—(1) modern humans, (2) Neanderthals, (3) Denisova hominins (Krause et al. 2010), and (4) *Homo floresiensis*—occupied different parts of the globe, the task of tracking and examining the patterns of Pleistocene hominin evolution has become more complex and more interesting. How did we go from this diversified state with multiple taxa to the current situation, in which only modern *Homo sapiens* are alive today?

If we define all living people, despite their vast behavioral variation, as "modern" (Antweiler 2007), it is legitimate to ask when in the course of human evolution people became modern. This question is certainly flawed for reasons Shea points out, but it is still a valid question that helps us to define the human condition and our place in the diversity of life. If archaeologists and paleoanthropologists, who produce and study the key data, sidestep this issue, we leave this job to the practitioners of other fields, who are almost always farther removed from the data and less competent to answer this question. Many scientific questions are flawed, but that does not mean that all flawed questions are unimportant or inappropriate topics of scientific inquiry. Like most concepts, Shea's emphasis on the study of behavioral variability is also flawed, as well as difficult to implement.

Shea attempts to shift the debate away from indicators of "modernity" and toward an analysis of variation in cultural systems. This, he suggests, should be done using models based on evolutionary theory and, more specifically, behavioral ecology. This position is by no means new, and Shea is preaching to the choir on this point, because most researchers today regularly use concepts from behavioral ecology. The issue, as Bird and O'Connell (2006) and many other researchers have

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pointed out for years, is how can we best do this? If developing methods to conduct these kinds of studies were easy, we would already have many convincing examples of such studies rather than continued calls to develop ways to proceed along these lines.

While Shea rejects the notion of behavioral modernity, he exchanges this concept for the equally vague concept of "current degree of behavioral variability." His article argues that the presence and relatively heavy reduction of red jasper and the presence of nonutilitarian greenish white opal silica from at least 15 km away from its place of origin in the late Middle Pleistocene of East African plausibly document symbolic behavior. This argument strikes me as inconclusive and not so very different from what Shea criticizes as "cherry-picking" in other contexts. The discussion of variation in the use of Clark's modes in lithic assemblages is equally vague and inconclusive. As Shea writes, multiple technological modes often coexist in archaeological contexts. This is neither new nor exceptional.

Shea also claims that the concept of behavioral modernity raises problems related to what he refers to as "recursive" behavior. He states that this problem can be avoided by a shift toward strategic modeling. I disagree. The way cultural patterns and kinds of artifacts come and go requires explanation, regardless of which perspective one uses to examine temporal changes in technology. It is only the stereotypical, strawman models that Shea portrays that rely on notions of unilinear evolution and an essentialist dichotomy between binary states of modern and nonmodern. Very few researchers I know approach these issues from such schematic points of view.

I agree fully with Shea that we need testable models that can be operationalized using the archaeological record. For some years I have been conducting excavations to gain high-quality data from several regions to try to address these questions and formulate testable hypotheses (Conard and Bolus 2003; Conard et al. 2006). One promising approach is to examine under what specific settings one hominin was able to outcompete another (Conard 2008, 2010). This is closely related to the question of why it took anatomically modern humans at least 150,000 years to replace Neanderthals and other archaic humans. Like Lewis-Williams (2002) and Shea (2003*b*), I argue that some cultural and technological innovations developed at the interface between different populations and contributed to the evolutionary success of one group. In other settings we can observe technological convergence and cultural exchange between populations. Recent studies by Green et al. (2010) have also shown that interbreeding between archaic and anatomically modern humans occurred. In my view the available evidence suggests that there was not one place and time where "modernity" evolved like the flipping of a switch. Instead we can view the evolution of behavioral patterns that fall within the range of living people as a polycentric, mosaic process, a diffuse membrane through which multiple populations passed in a variety of settings. This was not a one-time-only quantum leap but a process that took on different characteristics in different settings.

I welcome Shea's approach using cost-benefit studies and strategic modeling, but I also challenge him to give us better examples of how we can operationalize these ideas to formulate refutable models for the evolution of the current range of behavioral variability.

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The concepts of modern human behavior and/or behavioral modernity are loaded with overtones of evolutionary destiny. The call for a shift away from such terms is overdue, and this one is to be welcomed. Shea's article shows us how the quest for "modernity" brings little but harm to the study of human evolution, and he offers us a possibly viable alternative approach for the systematic analysis of Paleolithic behavior. Although more than willing to see a sea change in both the concepts and its terminology, I have to wonder if behavioral variability will be the panacea that Shea promises.

The principal challenge to this alternative approach will be methodological: how is "behavioral variability" to be measured? Why would this concept not be affected by all the same suspects that also plague behavioral modernity? To measure variability, some sort of units of input will have to be counted; once again it will be the researcher who decides which units are important enough to be included. This sounds suspiciously like the dreaded "trait list" so maligned by Shea. Preservation issues and sampling bias will continue to limit the categories of behavior that can be reached in the material record. The frequency and distribution of such behaviors, all required for a robust measure of variability, will continue to be haunted by the same false negatives he reviews here. Agreed, "behavioral variability" lacks the kismet stigma of its divisive predecessor, but it faces the same set of methodological challenges.

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Those interested in the origins or early evolutionary stages of our species, either anatomically or behaviorally, grapple with countless variables that may explain aspects of the preserved fossil and/or archaeological records. Adding to these chal-

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lenges is the inadequacy of our own constructs (e.g., trait lists) and terminologies. In this context, Shea unpacks issues regarding the use of the term "behavioral modernity" and the limitations of trait lists that aim to investigate such behavior. From the late 1970s onward, before the whining became fashionable, Africa-based archaeologists such as Hilary Deacon (sadly no longer among us; died May 2010) have questioned the usefulness of a Eurocentric yardstick for discussing the later stages of human evolution and cultural development on the continent (Deacon 1979:255). Thus, Shea will find sincere support for his endeavor to activate a constructive shift in how we approach the Paleolithic archaeological record as a source for studying aspects of human behavioral evolution.

This shift is already apparent in the work of researchers such as Langley et al. (2008), Wadley (2010*a*; Wadley, Hodgskiss, and Grant 2009), and Haidle (2010), among others, and will gain momentum as we proceed. For various reasons they are knowingly moving away from finding evidence for "behavioral modernity" or "fully symbolic behavior" toward methods that are able to assess levels of technological, behavioral, or cognitive complexity and/or variability. Generally, these approaches require long-term, multidisciplinary research programs dedicated to developing falsifiable hypotheses, suitable methodologies, fine-grained data sets, and reflective theoretical frameworks.

Scarce contextualized finds representing direct evidence for early symbolic behavior in Africa and elsewhere (e.g., Henshilwood et al. 2004; Vanhaeren et al. 2006) provided critical impetus for extending the search for behavioral modernity back in time. Yet I agree with Shea that it is time to cast a wider measurable interpretative net and to adapt our terminologies and definitions accordingly. "Working" definitions and terminologies have their place in the sequence of scholarly exploration; however, when cumulative evidence points to their growing deficits and ambiguity, it is time for revision. Simply put, such ambiguity or distortion can contribute to the misinterpretation of behavioral information contained in the Pleistocene record and to the perceived disparity between the fossil and archaeological records regarding the origins of *Homo sapiens* (e.g., Stern 2009).

I am optimistic about Shea's push to increasingly think about past human behavior in strategic terms. Such approaches, together with augmented empirical data, will reshape our understanding of humans who lived during the Pleistocene. On the other hand, I find his use of an adapted lithic modal framework as universal measure for behavioral variability among humans (e.g., first sentence of his conclusion) less persuasive. Shea's argument relies on the assumption that multiple modes of stone tool production within a single context equals behavioral variability. The data presented (table 5), are too coarse grained (most early contexts span many tens of millennia) to convincingly support his conclusion that the data show "that the earliest *Homo sapiens* populations who lived in East Africa around 200 kya possessed a capacity for behavioral variability *identical* [my emphasis] to 'behaviorally modern' Upper Paleolithic humans." It is impossible to know whether the modes ticked in table 5 can be associated with single cultural/behavioral contexts or whether they merely represent the accumulation of different knapping strategies over time.

Only fine-grained stratigraphic analyses (and data presentation) that are able to exclude any possibility of contextual mixing at the represented sites (if the geomorphology of the sites allow this) will be able to test the results presented here. The fact that few recent archaeological sites have a full lithic modal range may also contain a valuable clue regarding the aptness of this method for gauging broad-spectrum behavioral variability. This being said, I do not concur with those who reject stone tools and their associated technologies as important cognitive indicators. Stone tool assemblages from finely stratified, narrowly dated Middle Stone Age sites and the supplementary information they may hold have already produced unexpected behavioral resolution (e.g., Lombard 2008; Lombard and Phillipson 2010). These studies enable progressive cognitive modeling independent of evidence of so-called modern or symbolic behavior (Wadley 2010*a*; Wadley, Hodgskiss, and Grant 2009).

Lastly, I do not think the terms "modern behavior" or "behavioral modernity" will disappear from the literature any time soon. Whether for "careerist" reasons, lack of more appropriate terminology, or aiming texts at an already receptive readership, they have simply become too expedient for some to discard at once or completely. Nonetheless, I hope that this article will compel most researchers, who are currently producing knowledge about the behavioral evolution of our species, to consider them (use them or lose them) with awareness of the theoretical and interpretative shortfalls they contain.

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Shea's article is a welcome addition to the ongoing debate on modern human behavior. While I am in agreement with him concerning the shortcomings of the term "behavioral modernity," I am not convinced that a focus on "behavioral variability" is any less problematic. What is "more variable" or "less variable" is virtually impossible to quantify with any precision. If, as Shea suggests, we are relegated to using nominal-level data in the study of behavioral variability, then we are no further ahead than we were with studies of "modern behavior." In fact, researchers studying modern behavior already use nominal-level data either implicitly or explicitly—"symbol use" is not "present" or "absent," it is "ochre use," "burials," "personal adornment," "incised and notched objects" (e.g., McBrearty and Brooks 2000). Furthermore, a fo-

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cus on behavioral variability still means that researchers have to choose what variables to study (lithics, subsistence, land-use patterns) and across what categories ("modes," raw material, degree of specialization). These choices return us to the problem of compiling trait lists. But Shea's call for "research focused on the strategic underpinnings of human behavioral variability" is compatible with an emerging consensus within the field (Nowell 2010) that social, demographic, and cultural factors are key to understanding patterning and variability in modern behavior as seen through the archaeological record. It is within this context that I will briefly address one of Shea's arguments.

Shea writes that "nearly all current models of modern human origins share the conviction that there are significant behavioral differences between the earliest *Homo sapiens* and those extant since 40–50 kya." He further observes that the pulses of modern behavior that characterize some regions, what he refers to as "recursions," require researchers to explain the "loss" and "re-evolution" of this capacity—a problem obviated by a focus on behavioral variability. In reality, only a minority of researchers—for example, Klein (2000, 2003, 2008) and Mellars (2007)—make a distinction between *H. sapiens* before and following 40–50 kya. If, following Marean (2007) and others (e.g., Chase 2003; Davidson and Noble 1989; Wadley 2001), we agree that modern behavior has symbol use at its core, it becomes increasingly difficult to explain why the earliest AMH populations—such as those represented by Qafzeh, Skhul, Blombos, Diepkloof Rock Shelter, and Pinnacle Point—are not behaviorally modern. Klein's (2008) model, for instance, would not recognize the engraved pieces of ochre and perforated tickshells from Blombos (d'Errico et al. 2005; Henshilwood, d'Errico, and Watts 2009) and the engraved ostrich eggshell containers from Diepkloof Rock Shelter (Texier et al. 2010) as evidence of modern behavior. Both are MSA sites with relevant artifacts dating to 77,000–70,000 BP and 65,000–55,000 BP, respectively. The latter example is all the more ironic given that Klein is an author on the Diepkloof Rock Shelter article. In that article, the authors write that the engraved ostrich eggshell containers "provide unequivocal evidence for the existence of symbolically mediated social behavior and the development of a behaviorally modern system by 60,000 years ago" (Texier et al. 2010:6184).

Shea asks how broadly we can extrapolate "a capacity for a particular behavior from a small number of archaeological occurrences." I would argue that once a capacity evolves in anatomically modern humans, we can infer its presence in all members of this species, extinct or extant. In fact, there is a growing consensus in the field that once the capacity for "modern behavior" evolves, its expression, that is, its appearance and disappearance at different times and in different places, is best explained with reference to social, cultural, historical, and demographic factors instead of cognitive ones (e.g., Barham 2007; Bar-Yosef and Kuhn 1999; Brumm and Moore 2005; Chase 1999, 2001, 2006; Conard 2006; d'Errico 2007; Habgood and Franklin 2008; Henry et al. 2004; Hockett and Haws 2005; Hovers and Belfer-Cohen 2006; Kuhn and Stiner 2007; McBrearty 2007; McBrearty and Brooks 2000; O'Connell 2006; Powell, Shennan, and Thomas 2009; Shennan 2001; Soffer 2009; Straus 2009; Wolpoff et al. 2004; Zilhão et al. 2010). Therefore, Shea's argument that the observed "recursions" require some extraordinary explanation is based on a false assumption—we are not talking about the "loss" and "re-evolution" of capacity, merely the differential expression of this capacity in an archaeologically visible fashion in relation to circumstance. This emphasis on the modification of behavior in relation to circumstance is not unlike Shea's call for a focus on the "strategic underpinnings" of human behavioral variability.

As Straus writes (2009:10), variation is "a phenomenon to be studied in order to get at the complexity of hominid behavior and adaptations. Just as the MP increasingly shows signs of 'modernity' so too would a 'noisier' UP come closer to reflecting the diversity of human strategies." Thus, I agree with Shea that behavioral variability is key to understanding the past but only as a vehicle for studying behavioral modernity and not as an end unto itself. This research must be undertaken within a robust theoretical framework that will allow us to assign meaning to the variation we document.

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Shea's article makes two important points: (1) conventionally identified archaeological indicators of behavioral capabilities typical of recent *Homo sapiens* are unreliable, and (2) variation in the distribution of those "indicators" across much of the Eastern Hemisphere over the last 200,000 years may have been determined primarily by ecological factors. I agree.

Shea cites archaeological data from Sahul (Pleistocene Australia–New Guinea) in support of his first point but underplays their nature and relevance. This region was first occupied by *H. sapiens* about 45,000 years ago (O'Connell and Allen 2004). Genetic data show that the people involved originated in East Africa and were closely related to those moving into western Eurasia at about the same time (Hudjashov et al. 2007; Joris and Street 2008). Colonization required the use of watercraft capable of open-ocean voyages greater than 100 km in length (Birdsell 1977). Demographic modeling and mtDNA data indicate that founding populations were substantial: minimum effective sizes in the scores, minimum census sizes in the hundreds (O'Connell, Allen, and Hawkes 2010). Colonizers reached locations thousands of kilometers beyond the initial entry point(s) within centuries. People in some areas maintained open-ocean voyaging capabilities and economies based in part on pelagic fishing at least intermit-

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tently to the ethnographic present (Allen and O'Connell 2008; O'Connell, Allen, and Hawkes 2010).

These achievements notwithstanding, the archaeological record of Sahul displays few of the commonly nominated markers of behavioral modernity through its first 25,000 years. The more than 70 pre-LGM sites so far reported have yielded a scant double handful of shell beads, one case of human cremation, and limited (if any) evidence of rock art (O'Connell and Allen 2007). Bone tools are uncommon and where present are simple in form and function. Evidence of the use of ochre is widespread, but its implications with respect to "modernity" are uncertain, given its recurrent presence in sites created by putatively "nonmodern" Neanderthals. The bulk of the record is made up of simple stone tools, generally unmodified or unifacially flaked and formally unpatterned except as a function of the materials from which they were made and the mechanical requirements of the uses to which they were put. In short, this record is much simpler overall than that of the European Upper Paleolithic, the archaeological baseline on which behavioral modernity is commonly assessed. "Standard indicators" of that quality become common in Sahul only in the Holocene. On those grounds, early Sahul indigenes were not "behaviorally modern" (cf. Balme et al. 2009; Habgood and Franklin 2008).

Many Paleolithic archaeologists ignore this inconsistency in favor of continued reliance on "standard indicators" (Klein 2008). Others attribute it to a shrinking of the collective Sahul mind induced by genetic and demographic bottlenecks that developed on route from Africa (Mellars 2006*a*). Still others see it as a function of sedimentary environments peculiar to Sahul and hostile to the preservation of organic materials of which many "standard indicators" were fashioned.

The limits of the first reaction are obvious. The second begs questions about possible bottlenecks encountered en route from Africa to Europe. It also fails to account for the increasing complexity of many aspects of the Australia–New Guinea record from post-LGM times onward, including an array of innovations similar to those noted on other continents across the same time span (notably the development of agriculture), most generated independent of outside influence by the descendants of those same "cognitively impaired" early colonists. The third observation may account for some of the differences between the Sahul and European records, but it cannot explain the contrast in lithics, a key issue in that many Paleolithic specialists see complexity in this domain—and equally important, its absence—as a fundamental index of human cognitive capabilities. The problem with all of these arguments lies in identifying the European Upper Paleolithic as the basic archaeological index of "true" humanity. As the Sahul record clearly shows, that proposition is untenable.

What now? Tracking the origins of modern human capabilities archaeologically is likely to be very difficult, especially in the absence of a theoretical framework that stipulates the circumstances that might have led to their emergence and that identifies potential archaeological tests. As I see it, that framework has not yet been developed. If, on the other hand, the goal is to explain variation in the archaeological record of Pleistocene *H. sapiens*, then the outlook is far more promising. As Shea says, the theoretical framework of behavioral ecology offers a proven approach to addressing this question. It has been applied successfully to a wide range of archaeological problems in the Holocene (Bird and O'Connell 2006) and explored in more limited but still promising ways (referenced but not much discussed by Shea) in connection with earlier *H. sapiens* (e.g., Henshilwood and Marean 2003; O'Connell 2006; Stiner et al. 2000). Making more comprehensive use of it should yield interesting and important results.

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Homo sapiens is a 10? John Shea argues that the concept of behavioral variability offers a far more useful quantifiable approach to understanding the evolution of human behavior than the dominant paradigm of behavioral modernity. His article provides compelling reasons why the current paradigm needs to be supplanted. My concern is that the "demonstration" of behavioral variability depends on detecting the presence of Grahame Clark's modes 1, 2, 3, and 4 in the early archaeological record of African *H. sapiens*. A cynical view, perhaps, is that from the time of its first appearance datum (FAD), if not before, *H. sapiens* represents a 10 on this measure of behavioral variability ($1 + 2 + 3 + 4 = 10$), with the different modes expressed at various times and places depending on the circumstances. I will expand on this point, but I will first note why it is wise to embrace the intent of Shea's argument.

The article builds on evidence established by archaeologists who work on the African mid- and late Pleistocene. This growing body of research shows an earlier expression of behaviors commonly associated with *H. sapiens* than is encountered in the European archaeological record. The paradigm breaker that Shea proposes involves avoiding trait lists intended to define when archaeologically inferred behaviors can be placed in the "modern" category. Because of preservation biases, ongoing revision in the archaeological FADs for particular behaviors, and other factors, Shea notes that the overall package of "modern behavior" will mistakenly look as if it had emerged incrementally within *H. sapiens*. Some researchers, therefore, detect a threshold increment in "modern human traits" in Europe nearly 50,000 years ago, while others stress a longer emergence of "modernity," commencing with the origination of *H. sapiens* or somewhat before. Thus, according to Shea, we are stuck in a mire of conceptually

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stagnant possibilities, none of which matches the philosophically powerful (but is it correct?) perspective that all *H. sapiens* back to roughly 200 kya must be equal in their inherent adaptive possibilities.

The reason to embrace the paper's main thrust is that Shea is on the mark in his tough critique of progress-oriented paradigms, limits of extrapolation, and the difficulties in pinpointing the actual origination time of any particular behavior. He gives ample reasons to look for a novel analytical approach and terminology to supplant the concept of behavioral modernity.

The paper thus leads one to expect that a quantitative statistical analysis will capture measurable dimensions of behavioral variability and will help us understand and possibly even better explain the evolution of human behavior. More than one reader will be surprised, then, that Shea's principal analysis invokes Clark's system of technological modes to show that East African lithic assemblages from roughly 300 to 100 kya possess the "type artifacts" of modes 1 through 4. Shea takes this as a measure of wide behavioral variability in *H. sapiens* from its onset. This is surprising, because Clark's system typifies the linear evolutionary paradigm that Shea decries. It is also surprising because the presence of these modes is considered a demonstration of behavioral variability somehow related to the claim that it is a measurable, statistically testable concept. Finally, it is surprising because, while *H. sapiens* could always express four technological modes or any one of them depending on the circumstances, it implies that Acheulean handaxe makers only had modes 1 and 2 at their disposal while some later hominins had three of the modes at their disposal. We end up, therefore, with a linear incremental account of the evolution of human behavioral variability. This particular comment misrepresents Shea's admirable intention, but I do not think it misrepresents the poor choice of the modal system in analyzing the problem. No viable reason is given as to why the accumulation of these four technological modes should reflect the strategic versatility of *H. sapiens* in response to diverse contexts.

The point of this critique is that Shea's paper aims so impressively for a conceptual breakthrough that I really wished to see a quantifiable statistical treatment of behavioral variability, one that would match the expectation. It is yet to come. Seeking to augment his argument, Shea then highlights the possibility of symbolic behavior in the lithic materials from Omo Kibish, where the current morphological FAD for *H. sapiens* occurs at ~195 kya. As acknowledged, however, this avenue is akin to using a trait to locate behavioral modernity in the archaeological record. The case for behavioral variability is not helped.

In the end, the article made me wonder whether behavioral variability is an explanatory concept that could improve paleoanthropological analysis or whether it is, instead, something that requires an explanation. If it is the latter, the explanation will likely be based in an analysis of evolutionary costs benefits and how adaptability evolved in hominins, enabling successful responses to varied adaptive settings. Shea's bark in that direction draws attention to the right tree.

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10 VIII 10

John Shea contributes a welcome and wide-ranging essay on the importance of Paleolithic archaeology for understanding the behavioral evolution of *Homo sapiens*, building on earlier work that also focused on Africa by McBrearty and Brooks (2000) and Henshilwood and Marean (2003), among others. Shea argues for abandoning the concept of behavioral modernity and the search for a single all-encompassing measure of "modern" and (by default) "nonmodern" behavior in the archaeological record during the period following the first appearance of *H. sapiens*. He instead advocates an emphasis on the very human characteristic of behavioral variability and an understanding of the costs and benefits that may explain patterns in the "strategic choices" made by Pleistocene hominins. This latter suggestion follows along the lines of research approaches practiced by, among many others, Binford (2001) and Stiner and Kuhn (2006), and it is likely to yield further important insights. Much of the theory in this article is sound and will provide talking points for some time to come.

Clearly the challenge lies in tenable tests of human behavioral variability. As a test of his core hypothesis of equivalent behavioral variability throughout the history of *H. sapiens*, Shea draws on lithic artifact data from Middle Pleistocene–Holocene archaeological sites in northern Kenya and Ethiopia using Grahame Clark's (1969) technological modes. I would argue that there are better tools for measuring variability. As a normative typological device, the Clark system drastically simplifies the complex decisions and behaviors that went into stone toolmaking, use, and discard. Given this limited range of variation, that broad similarities exist is not surprising. On the other hand, if other more nuanced measures of comparison among lithic assemblages, such as those suggested by Tostevin (2007), were also to yield similarities, a stronger case for equivalent behavioral variability would be made.

As in all such comparisons, temporal resolution among the sites also raises some challenges. As Shea recognizes, larger samples (accumulated over more time) are more likely to produce more variable artifact assemblages. Extensive refits from sites in the Kibish Formation (Sisk and Shea 2008) suggest the rare preservation of brief "snapshots" of the past there and archaeological variation within these assemblages may be reasonable measures of the behavioral repertoire of the population that made them. The other site complexes in table 5, however, are time averaged to varying degrees, combining in some instances artifacts from numerous sites and

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multiple strata likely deposited over thousands of years and across hundreds of generations. Potentially, the degree of variation among the site complexes is confounded by differences in their temporal scale. One possible solution would be to limit comparison to individual assemblages with high temporal resolution (as suggested by multiple refitted sets, spatial analyses, or geological observations), including KHS and BNS from the Kibish Formation, Locus 1 at Koimilot in the Kapthurin Formation (Tryon 2010:52–53), ETH-72-8B at Gademotta (e.g., Wendorf and Schild 1993:41), and the deltaic sands (level 3) of the Middle Excavation at Lothagam (Robbins 1974).

Such a scale-adjusted set of assemblages would offer information about the variability of hominin behavior at different times and in different places, yet for much of the Pleistocene, these data are not likely to speak to differences among species. At any point in time, the archaeological record is frustratingly mute on the identity of its makers, but the situation is particularly acute during the Middle to Late Pleistocene. While our understanding of the first appearance datum for *H. sapiens* has increased substantially over the last decade, the same cannot be said for ancestral or sister taxa in Africa during the later portion of the Middle Pleistocene. If we assume a cladogenetic mode of speciation, then at the continental and likely at the regional level, two or probably more very similar hominin species contributed to the archaeological record during the period under consideration. That all African sites less than 200 kya are the products of *H. sapiens* is therefore an untested (and I would suggest, unnecessary) assumption.

Instead, I would argue that we need to adopt the population (rather than the species) as our basic unit of analysis, a scale commensurate with documented regional differences in the archaeological record (Clark 1988) and recent analyses of the genetic and human fossil data (e.g., Crevecoeur et al. 2009; Gunz et al. 2009; Howell 1999; Lahr and Foley 1998; Pearson 2008). Understanding the relative contribution of social and environmental factors (as opposed to biological ones) to archaeological variability—whether a result of adaptation, agency, or drift—requires a highly resolved record of artifacts and paleohabitats within a sound chronological framework of the sort recently done in the Omo Kibish region of Ethiopia (e.g., Fleagle et al. 2008). Clearly, however, our work is cut out for us in order to build such a comparative framework across the rest of Africa.

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The terms "behavioral modernity" and "modern human behavior" lack analytical precision and theoretical grounding, and as John Shea argues, they need to be dropped. I share Shea's belief that a productive alternative to searching for elusive "modern behavior" is to examine the evolution of particular behaviors and their variability in the deep past. Of course such research needs theoretical underpinning if it is to be successful. One example of why this is necessary comes from the attempt to study one particular behavior, the evolution of language. The less than 70 kya shell beads from Blombos Cave have been used persuasively by influential archaeologists as evidence for fully syntactical language, yet Botha (2008) contends that this conclusion at present lacks the necessary inferential stages. He claims that the steps leading to an interpretation of language evolution need to be suitably "grounded" and "warranted." Grounding demands empirical evidence for an inference; warranting stipulates that the evidence must be appropriate for connecting the data to the inference (Botha 2008; Wynn 2009). It is necessary to use "bridging theory" to span the gap between "departure" phenomena and "destination" phenomena (Botha 2008) so that a sequence of arguments persuasively links the archaeological data to the interpretation. Archaeological inferences are only as strong as the bridging arguments on which they depend; these need to be well grounded, and the data used must be pertinent to the eventual conclusions. One of the reasons for the failure of "modern behavior" interpretations is that they sidestepped theoretical bridges and made a leap of faith between items of material culture and evaluations of cognitive capacity. This resulted, among other things, in assuming causative links between skillfully produced bone or stone tools and "modern behavior" and between the presence of ochre pieces and symbolic behavior. In the absence of bridging theory, such inferences are weak.

Ten years ago I thought that researching exosomatic symbolism was the best way to find out when people started to behave somewhat "like us" (Wadley 2001); today I am less certain about this possibility. As Shea points out, symbolic behavior is as difficult to define satisfactorily as "modern behavior," and there is no consensus about its definition. In addition, symbolic behavior is sometimes inferred from archaeological data without consideration of the necessary mental architecture. Like Shea, I am now inclined to shift my attention to the evolution of technology. However, Shea seems skeptical about the possibility that "tasks that require the conjunction of several disparate components . . . might be connected by some shared cognitive capacity." I feel less pessimistic than Shea about this aspect of cognitive research and am convinced that a careful study of technology can yield information about the artisans' cognitive status. If some aspects of a technical process cannot proceed without cognitive attributes that imply complex cognition, then the technology in question is informative about cognitive capacity. Our own cognitive abilities grant us, among other things, abstract thought, multitasking, and mental flexibility that enables us to switch attention between tasks. Such mental abilities are

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inferred, for example, by the ninth subsystem of Barnard's (2010) cognitive model of mental architecture. Here, abstract meanings and the organization of action sequences determine decision making. Within Barnard's model, an earlier eighth subsystem implies that the incumbent hominid was mentally incapable of either processing two levels of meaning simultaneously or of generating abstract concepts about behavior.

Through replications, it is sometimes possible to trace the cognitive steps involved in the manufacture or use of items of material culture. Now and again these steps (such as those required for the manufacture and use of compound glues that combine several ingredients, procedures, and complicated pyrotechnology) cannot be executed without recourse to abilities such as abstract thought and multitasking (Wadley, Hodgskiss, and Grant 2009). In my view, this suggests overlap between the cognitive abilities of people living today and people who made compound adhesives in the Middle Stone Age or the Middle Paleolithic. The concept of remote capture involved in the creation and use of snares and traps seems to be another indicator of enhanced working memory and complex cognition (Wadley 2010*b*). Equipment designed to function out of sight of its maker and not immediately, but at a future time, provides evidence for an ability to integrate action across space and through time (Wynn and Coolidge 2003). This ability engages modern executive functions of the brain that in turn characterize enhanced working memory (Coolidge and Wynn 2005; Wynn and Coolidge 2003, 2007) and hence complex cognition. The central executive is the decision-making component of working memory, and its functions include paying attention to the goals of a task at hand and inhibiting extraneous thought and action (Wynn and Coolidge 2007). The use of models such as the Working Memory Model as bridging theory (in the sense used by Botha 2008, described above) enables us to link technology such as snaring with modern executive functions of the brain (Wadley 2010*b*).

Shea has a superb knowledge of lithic technology and also has the ability to replicate a variety of stone tools. I feel confident that, should he wish to do so, he would be able to trace cognitive development through the evolution of lithic technology.

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Based on analyses of the Eurasian evidence, for the past 15 years I have argued at length that the concept of behavioral modernity is flawed and misleading (e.g., Zilhão 2006*a*). Therefore, I cannot but welcome Shea's basic contention—that the concept should be abandoned.

I note, however, that Shea falls short of recognizing that the real problem with behavioral modernity lies not in its failure to discriminate in archaeologically or evolutionarily meaningful ways between different groups of Paleolithic humans but in the fact that the concept is inextricably linked to a wider paradigmatic construct whose main tenets are that (*a*) different biological species of *Homo* coexisted in the Middle and the Upper Pleistocene, (*b*) by definition, each of those species was characterized by a species-specific behavior as much as by a species-specific morphology, and therefore (*c*) a "modern behavior" exists that is exclusive to modern humans (or *H. sapiens*) and, accordingly, has not (and could not possibly have) left any manifestations in the archaeological record of nonmodern ones (e.g., the Neanderthals or *Homo neanderthalensis*). It is that paradigm, the "human revolution," that needs to be abandoned, because despite Shea's *passim* claims to the contrary, it is now established beyond reasonable doubt that (*a*) even by criteria derived from the European Upper Paleolithic, Neanderthals were behaviorally modern (d'Errico 2003; Zilhão 2001; Zilhão et al. 2010); (*b*) Neanderthals interbred with modern humans and made a significant genetic contribution to the genome of present-day people (Green et al. 2010; Trinkaus 2007); and therefore (*c*) Middle and Upper Pleistocene fossil "types" are expressions of the interpopulation morphological variation that existed within a single evolving human species. That the "human revolution" fails to explain the evidence even in Africa is not irrelevant but, in the overall scheme of things, little more than an interesting detail.

One of the most deleterious effects of the "human revolution" has been the trend to reduce research largely to the measurement of the extent to which past human societies fitted one of the two dichotomically opposed categories of "modern" or "archaic." In this framework, the explanation of modern behavior then became the tautological reiteration that observing the corresponding features in the archaeological record proved that (*a*) modern humans did behave as modern humans and (*b*) modern humans behaved as modern humans because they had the capability to behave as modern humans. Although Shea blames European archaeology and Eurocentric views of the archaeological record for this state of affairs, the truth is that it mostly results from the efforts of Africanists to reconcile the concept of a "human revolution" with the realities of their record. In fact, for well more than a decade now, archaeological research on the Middle-to-Upper Paleolithic transition in Eurasia has been able to bypass the "human revolution" dead end and move on, in practice if not in theory, toward illustrating and explaining the variation in the cultures and adaptations of the time period, focusing on the understanding of societies per se, not against an abstract standard of "modernity" (e.g., Hopkinson 2004; Richter 2000; Vanhaeren and d'Errico 2006). Shea's call is one for Africanists to join in, and one can only hope that he will be heard.

Shea further suggests that explanations of human behavioral variability should be grounded in cost-benefit analysis.

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That would certainly represent a methodological improvement over searching for the genetic switch whose turning on would have triggered the emergence of an immanent modernity. Such analyses are unquestionably useful when it comes to setting the limits of the possible, but one must bear in mind that they are prone to the same fundamental error of biologically based approaches—that of operating in a historical vacuum, as if "behaviorally" (i.e., climate and environment being equal), a person or society of 100,000, 10,000, or 1,000 years ago was subjected to essentially the same time/energy constraints. This is not the case, because human adaptations are mediated by technology and social organization, and cumulative developments in extractive efficiency (i.e., the invention of better tools), in the division of labor, and in the networking of individuals and groups may conspire to bring about a "release from necessity" that escapes cost-benefit analysis. For instance, obtaining a certain raw material of particular quality may be very expensive if you have to walk 300 km to get it yourself but fairly inexpensive if it can routinely arrive at your doorstep through down-the-line exchange systems that increase the productivity of the land via mechanisms of group specialization that take advantage of the patchiness of resources and bring about the higher population densities required for the exchange system to work.

In short, there is more to the Aranda proverb quoted by Shea ("The more you know, the less you need") than meets his cost-benefit eye. That is why (and especially when dealing with long-duration issues) Paleolithic archaeology would fare much better if it understood that primarily, it is all about (pre-)history.

Reply

I thank my colleagues for their insightful comments. There appears to be a consensus that behavioral modernity is a problematic concept in human origins research. Yet there are also differing views about how this problem originated and whether it is of global or merely regional significance (Nowell 2010). The core argument in this article is that there are no such things as modern humans, no such thing as behavioral modernity, and that retaining these concepts in Paleolithic archaeology is doing more harm than good. They are spreading to other behavior science fields (e.g., Fuentes 2009), and after this happens, it will take a lot of time and trouble to root them out. Like "race," behavioral modernity is a legacy of typological thinking about human variation. Paleolithic archaeology, paleoanthropology, and anthropology as a whole are all better off without behavioral modernity.

I wrote this article because I found myself dealing with the archaeological record for early (pre-50 kya) *Homo sapiens* in two very different contexts, the East Mediterranean Levant (Shea 1989, 2003*a*, 2003*b*, 2007*a*, 2008*b*, 2010) and the Lower Omo Valley, Ethiopia (Shea 2008*a*; Shea, Fleagle, and Assefa 2007). Much of what has been written by archaeologists about modern human origins over the last 40 years asserts that the behavior of these early *H. sapiens* differed from the behavior of humans who lived after 50 kya for deep-seated biologically based reasons. I even believed this myself for a time (Shea and Bar-Yosef 2005) until I recognized that it was unscientific to accept such an argument without having tried to falsify the null hypothesis of "no difference." There are differences between pre- and post-50 kya humans, to be sure (Shea 2007*a*), and important ones at that (Shea and Sisk 2010), but splitting our species into "archaic" and "modern" humans does not move us closer to understanding those particular differences or the sources of their variability.

With this article, I challenge Paleolithic archaeologists either to discard behavioral modernity as an analytical construct or to come up with a clear and convincing case for continuing to use it. Among the comments, I find no principled argument for retaining behavioral modernity in human origins research. (It functions well as bait for journalists, but that's about it.) The only real resistance I sense is from researchers for whom behavioral modernity remains a convenient shorthand for symbol use, language, and complex cognition (e.g., Henshilwood and Marean 2003; Nowell 2010; Wadley, Hodgskiss, and Grant 2009). This is fine, but would it not be easier to just speak about symbol use, language, and complex cognition rather than bundling them together? Symbolism and language are among our species' important derived characteristics, but it is a mistake to think one can rescue behavioral modernity by distilling *H. sapiens* uniqueness down to symbolic behavior. The presence of the FOXP2 "language" gene in aDNA from Neanderthal fossils shows that there was strong selective pressure for spoken language, and thus prior selective pressure for symbolic communication, among our last common ancestor if not among still earlier hominins. *Homo sapiens* was probably not the only linguistic, symbol-using, cognitively complex hominin. We are just the only one who survives to the present day.

There is more to human uniqueness than symbol use, language, and complex cognition. *Homo sapiens* differs from our near primate relatives in a variety of interesting ways. If we are looking for root causes of evolutionary change, the first place to look is among variation in reproductive strategies (Darwin 1859). For example, *H. sapiens*' reproductive strategies involving pooled energy sources effectively balance slow child development with rapid birth rates (Kramer and Ellison 2010). These qualities almost certainly underwrote our species' population growth, rapid geographic dispersals, construction of broad ecological niches, development of complex technology, and reliance on extensive social networks. While language, symbolism, and complex cognition might have played a role in the institution of such reproductive strategies, they are far more likely epiphenomena developed in service of those strategies. It will be difficult for us as Paleolithic archaeologists to investigate variation in human reproductive

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strategies with the kinds of data we are accustomed to discussing. Nevertheless, this is a challenge for future research, not a known impossibility. That speech does not fossilize any better than sex does has not stopped archaeologists from developing creative ways to investigate language.

Several commentators found my use of Grahame Clark's five technological "modes" to describe variability in the East African evidence problematic. They were not my first choice either. I used Clark's modal framework for two reasons. First, if one wants to show that current approaches to the Paleolithic record are insufficient, then one has to showcase these problems in familiar terms. Clark's framework is in broad usage among prehistorians working in Africa and Eurasia (Barham and Mitchell 2008; Dennell 1983; Foley and Lahr 1997; Gamble 1999; Norton and Jin 2009). Second, it was possible to assess from published records whether or not East African lithic assemblages featured the defining characteristics of each mode. This is no small feat. Eastern Africa is a region of typological anarchy. If mode 4 = Upper Paleolithic = modern human behavior/behavioral variability, then insofar as the lithic evidence is concerned, the degree of human behavioral variability present in the later Middle Pleistocene of eastern Africa is equivalent to that seen in the European Upper Paleolithic. The null hypothesis stands. The phrase "insofar as the lithic evidence is concerned" of the penultimate sentence is crucial. This analysis focused on the broad technological characteristics of the lithic record because that evidence is often used as a proxy measurement for prehistoric human abilities and activities. A survey of zooarchaeological evidence, settlement patterns, or even some other dimension of lithic variability might yield different results. Would I recommend using this approach to assess behavioral variability in other regions and time periods? Absolutely not. Pairing nominal scale modes and dichotomous (+/−) measurements is the least nuanced way of characterizing variability in any phenomenon. Circumstance may force one to use this approach (as it did in this case), but multivariate analysis of ratio-scale variables is assuredly a better way to go (see below).

Several commentators also expressed a sentiment that a research agenda focusing on prehistoric human behavioral variability might be not be that much different from studies of behavioral modernity, or that it might be a flawed and unmanageable analytical goal. Simply rebranding behavioral modernity as behavioral variability will be no more successful in stimulating new research than was relabeling creationism as intelligent design. But as Adler, Conard, and others note, archaeologists have been conducting studies of prehistoric human behavioral variability in terms of strategic modeling and behavioral ecological principles for quite some time now. Data from Pleistocene contexts are fewer and more sparsely distributed than those available to archaeologists investigating human behavioral ecology in Holocene contexts, but this is not a serious obstacle to testing hypotheses derived from behavioral ecological models. One simply has to adjust one's thinking about variation from historical timescales to geological ones.

This article was not intended to be an infomercial for any particular approach to studying behavioral variability in the Paleolithic record. Nevertheless, as Potts remarks, it would have been better to show an application of the research strategy I advocate. Two applications of such an approach explicitly modeling of costs and benefits associated with stone tool technology have already been published, but these are embedded in longer descriptive works (Ahmad and Shea 2009; Shea 2008*a*; Shea, Fleagle, and Assefa 2007). Therefore, key elements of this approach are outlined below.

In behavioral ecological modeling, one has to determine the costs and benefits at issue. For example, flakes are the most common kind of artifact in most lithic assemblages. Flakes differ from other lithic artifacts in the same assemblages as well as from stone tools made and used by nonhuman primates in featuring sharp cutting edges. Patterns of retouch, microwear, and residues preserved on the edges of flakes indicate that these sharp edges were a desirable quality, something prehistoric knappers probably sought to optimize when they knapped cores from at least 2.6 Ma onward. Other qualities were undoubtedly sought as well, but the production of sharp cutting edge is the one shared characteristic of nearly all Pleistocene lithic technology. All other things being equal, flakes that provide more cutting edge (and/or potential cutting edge) represent a more successful effort to extract "benefit" from knapping activities than flakes that yield less cutting edge. For complete flakes, this property is easy enough to measure in terms of the ratio of flake surface area (technological length × midpoint width) divided by thickness at the midpoint of length (FSA/T; Davis and Shea 1998). These measurements are among the ones archaeologists routinely record for flakes (Dibble 1997). A flake with a higher FSA/T value is broad and thin, presenting more cutting edge (and more potential for curation by resharpening) than a short thick flake whose FSA/T value is relatively low. Patterned and significant increases FSA/T can be interpreted as efforts to optimize benefits in flake production.

If cutting edge is an identifiable benefit of flake production, what then is the currency of cost? Leroi-Gourhan (1964) proposed using mass as such a measure, but mass varies as a function of density among different lithic raw materials. Moreover, it is not a variable archaeologists routinely record for flakes. An alternative measure of cost involves metric variation in flake striking platforms. Beyond a certain minimum area necessary to allow conchoidal fracture initiation, every additional millimeter of striking platform width removes a portion of core edge that could otherwise have served as a launching point for an additional flake removal. The removal of this potential striking platform, therefore, is an identifiable cost to flake production. Flakes with a wider striking platform per unit of striking platform thickness (SPW/SPT) are more "costly" than those with a narrower striking platform (Davis

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and Shea 1998). Measurements of SPW and SPT are also routinely made on flakes from Paleolithic contexts.

Table 6 compares FSA/T and SPW/SPT values for whole flakes from three eastern African Middle Stone Age lithic assemblages in Members 1 and 3 of the Lower Omo Valley Kibish Formation (Shea 2008*a*). KHS and AHS are located in Kibish Member 1, which dates to 195 kya. KHS is the find spot of the Omo 1 hominin, and these lithics are associated with that individual (Leakey, Butzer, and Day 1969). AHS is a multilevel site at which a human tibia and fibula were recovered. BNS is located in a sandy deposit capping an erosional unconformity at the top of Member 2. The single-crystal argon date of 104 kya for Member 3 provides a minimum age estimate for BNS. Multiple sets of refitting stone artifacts and human fossils from KHS and BNS attest to the stratigraphic integrity of those contexts. The different layers of AHS are combined because no significant differences were found among them.

Table 6. Cost versus benefit in flake variation among Middle Stone Age assemblages from the Lower Omo Valley Kibish Formation

	KHS (n = 22)	AHS (n = 144)	BNS (n = 76)
Cost (SPW/SPT):			
Mean	3.43	4.1	3.9
SD	1.5	2.1	1.8
Variance	2.2	4.2	3.25
Benefit (FSA/T):			
Mean	171.4	201.0	159.6
SD	109.0	98.2	54.2
Variance	11,884.6	9,636.2	2,944.0

Note. KHS = Kamoya's Hominin Site; AHS = Awoke's Hominin Site; BNS = Bird Nest Site; SPW = striking platform width; SPT = striking platform thickness; FSA/T = flake surface area/thickness. n = sample size.

The relationships of cost and benefit in flake production among the Kibish assemblages are not simple. None of the differences in SPW/SPT, or cost, rise to the level of statistical significance ($P < .05$), though the differences between KHS and AHS come close ($t = 1.89$, df = 35, $P = .06$). The only significant difference in FSA/T is between AHS and BNS ($t = 4.03$, df = 218, $P < .01$). The older assemblages (KHS and AHS) cluster together in terms of somewhat greater variation in FSA/T but not in terms of variation in SPW/SPT. Plotted against FSA/T and SPW/SPT values for other assemblages (Ahmad and Shea 2009; Shea, Fleagle, and Assefa 2007), the Kibish assemblages exhibit widely variable values for cost but relatively low values for benefit. In fact, their nearest neighbors on that plot are samples of flakes knapped by *Homo erectus* 1.4 Ma at 'Ubeidiya in Israel. This analysis does not exhaust the Kibish assemblages' potential to shed light on early *H. sapiens* behavioral variability. It simply demonstrates that it is not all that difficult to construct cost-benefit models of hominin behavioral variability using lithic data we already routinely gather.

This cost-benefit approach to investigating behavioral variability with the lithic evidence will not be everyone's cup of tea. Nevertheless, if we want to study behavioral variability from a strategic ecological perspective, we have to do this in terms of costs and benefits. There are rewards for doing so. Frankly, these kinds of data potentially tells us a lot more about ecologically meaningfully variability in hominin behavior at these sites than scrutinizing them for evidence of behavioral modernity. Why are the benefit (FSA/T) values so low? Is this related to these sites' position near raw material sources? Did the Kibish knappers shift to more optimizing behavior over the course of lithic reduction (as Tryon notes, this is something that could be reconstructed from refitting sets of artifacts [Sisk and Shea 2008])? Are the benefit differences between AHS and BNS related to differing patterns of subsistence or ranging patterns? How does variation and variability in SPW/SPT and FSA/T among the Kibish assemblages compare with evidence from other Eastern African Middle Stone Age sites and to both Early and Later Stone Age sites in the same region? Does this variability match, overlap with, or exceed that in lithic assemblages associated with Eurasian hominins? Whatever the answers to these questions turn out to be, they will surely tell us more interesting things about the Omo Kibish people than whether or not they were behaviorally modern.

Different behaviors call for different ecological currencies and different kinds of archaeological data. It may not be possible to develop similarly quantitative and multivariate approaches to all aspects of the archaeological record. It is hard to imagine how one would go about calculating the cost and benefits associated with painting reindeer versus bison on the wall of a cave, but both activities probably have something to do with the energetic calculus of costly signaling (Bleige Bird and Smith 2005). Cost-benefit perspectives from diet breadth models can certainly shed light on why differing proportions of reindeer or bison (or birds, rabbits, tortoises, or shellfish) occur in Paleolithic faunal assemblages (Winterhalder 1986). Behavioral variability is, as Potts observes, not an explanation but rather something that once demonstrated requires explanation. Looking at behavioral variability in terms of strategic modeling of costs and benefits leads to far more interesting kinds of questions than those we have been asking about the Paleolithic record so far.

Much recent research on the behavior of early *H. sapiens* highlights "precocious" (i.e., early) occurrences of "modern" behaviors. Examples include color symbolism (Hovers et al. 2003), abstract symbolism (Texier et al. 2010), personal adornments (Henshilwood et al. 2004; Vanhaeren et al. 2006), pyrotechnology (Brown et al. 2009), complex projectile weaponry (Lombard and Phillipson 2010), and the exploitation of marine resources (Marean et al. 2007). These are all good studies by serious scientists. Nevertheless, one has to question what we are learning from such discoveries about the "big picture" of *H. sapiens* evolution. Does it really surprise anyone

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that skeletally modern-looking humans of Pleistocene antiquity exhibit the same range of behaviors as Holocene humans do? If it does, then one also has to reflect on the underlying assumptions about why such a finding is so surprising. Is it because we imagine people who look like us but do not act like us are something other than completely human? Is it because we are thinking of human evolution teleologically, as a process by which these "something other than modern" humans become "fully" modern humans? Neither of these assumptions withstands critical scrutiny.

A few other issues require brief comments. Barham and Zilhão argue that overlapping patterns of behavioral variability between *H. sapiens*, *Homo heidelbergensis*, and *Homo neanderthalensis* challenge the validity of those taxa. I disagree. If it is to be of any use whatsoever, hominin alpha taxonomy has to be based on morphological variation. Behavioral inferences are not as straightforwardly verifiable (or falsifiable) as observations of morphology even if the polarities of individual characters remain open to debate. Furthermore, behaviors can be acquired and transmitted between individuals in ways that genes cannot. That these three hominin taxa share many of the same derived behavioral characteristics simply tells us that we have not yet identified those behavioral characteristics responsible for their divergent evolutionary fates. Paleolithic archaeologists should not accept named hominin species as if they were actual self-conscious evolutionary actors any more so than we do the various named haplogroups molecular anthropologists envision scurrying around the Pleistocene world (Oppenheimer 2004; Sykes 2001). They are no less scientific constructs than named stone tool industries. The difference is that while we know species-level variation is a real thing among living organisms and that it is related to genetic and morphological variation, we do not know whether our arbitrary divisions of the archaeological record correspond to behaviorally, biologically, or evolutionarily significant divisions among the hominin populations who created them.

Several reviewers seize on recently published Neanderthal aDNA evidence to challenge the assertion that Neanderthals are irrelevant to debates about early *H. sapiens* behavioral evolution. These appeals are unconvincing. In fact, Green et al. (2010:722) raise and do not dismiss the hypothesis that the DNA in question is from an African common ancestor without living African descendants. I find this is a lot more plausible an explanation for the pan-Eurasian distribution of "Neanderthal DNA" than that there were sustained levels of interbreeding in Southwest Asia between fossil taxa that never appear together in the same stratigraphic context (Shea 2010). Whatever one thinks about this issue, it is simply a fact that during the period under discussion, 50–200 kya, *H. sapiens* was an African primate. Permanent *H. sapiens* dispersal to southern Asia long before 50 kya remains a possibility, but it is not one convincingly supported by fossil evidence (Shea 2008*b*). Neanderthals are fascinating creatures worthy of study in their own right regardless of their ancestral status. But unless we start finding Neanderthals in Africa or their DNA in either ancient or living Africans, they remain irrelevant for models of early *H. sapiens* behavioral evolution on that continent.

Like Conard, I am intrigued by the possibility that encounters between Neanderthals and *H. sapiens* populations in western Eurasia might have led to important behavioral changes among both species. To put hypotheses about such encounters in proper context, we need to know more about encounters between *H. sapiens* and other hominins in other parts of Eurasia and in Africa, too. If encounters with other hominin species stimulated behavioral innovations in *H. sapiens*, why was there no abrupt transformation of the archaeological record analogous to the European Upper Paleolithic on the island of Flores (Indonesia)? Why were the evolutionary consequences of encounters with *H. neanderthalensis* so momentous and those with *Homo floresiensis* apparently so inconsequential? (For *H. sapiens*, that is. For *H. floresiensis*, the consequence was extinction.) Why do we think (hope) analogous encounters in Europe had a different outcome? "Plays well with others" is not something one is likely to see on *H. sapiens*' evolutionary report card.

Wadley writes that as a flint knapper and lithic analyst I am well positioned to investigate the cognitive basis for Paleolithic stone tool technology. This is a flattering assessment, but I disagree with it. Flint knapping is no fast track to understanding prehistoric cognition. It is a fast track to the production of stone tools. I applaud colleagues who want to investigate the cognitive basis of lithic technology. To be done well, however, this kind of research requires the hard work of gaining expertise in cognitive psychology (e.g., Coolidge and Wynn 2009; see references cited in Nowell and Davidson 2010), just as gaining expertise in flint knapping requires an analogous level of hard work, if a somewhat larger supply of bandages.

Finally, Lombard may well be right that "behavioral modernity" is too well entrenched in the paleoanthropological literature to vanish in the near term. Yet people thought the same thing about "race" when Montagu (1945) led the attack against its use as an anthropological construct in the 1940s. It lingered for awhile, but nowadays one rarely sees "race" used in mainstream anthropological literature other than in reference to its historical role (Wolpoff and Caspari 1997). One can only hope that in years to come, readers of *Current Anthropology* will encounter behavioral modernity with diminishing frequency (so to speak).

—John J. Shea

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